



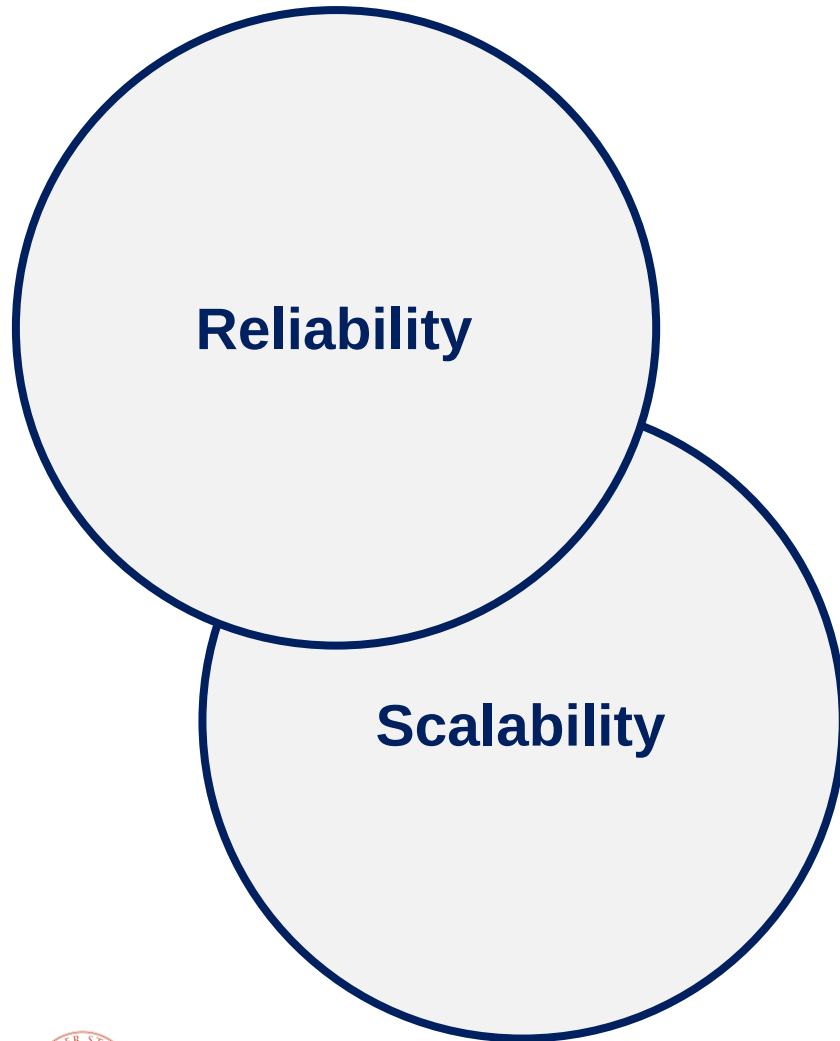
ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Content replication

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Replication: *not only the architecture*



Replication of server machines

Replication of interconnections

Replication of data

Replication of application processes

Replication of people



Data replication mechanisms

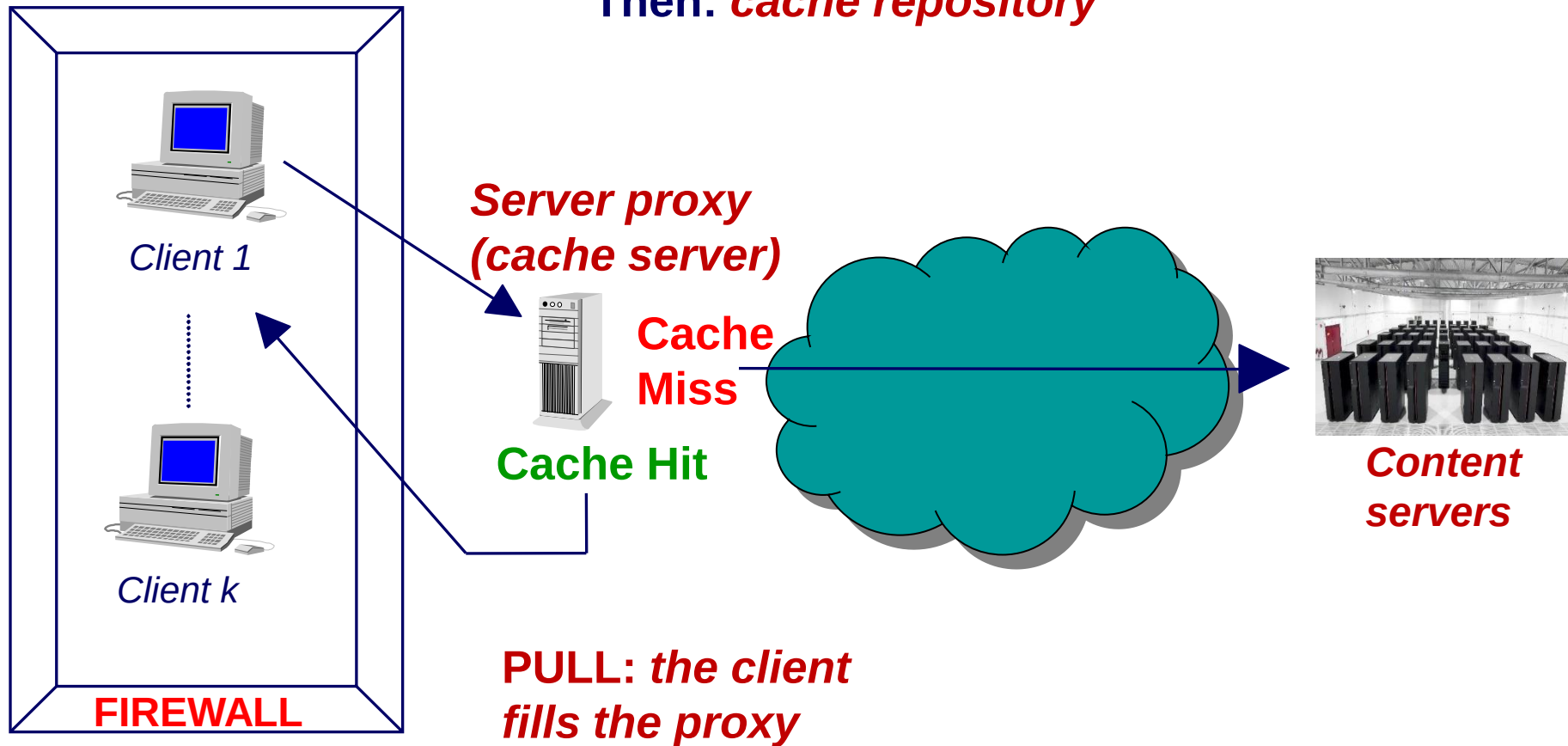
- **CONTENT CONSUMER**
 - Proxy server
- **CONTENT PROVIDER**
 - Data center + proxy servers
- **THIRD PARTIES**
 - ISP (*content consumer side*) → cooperative proxy servers
 - Companies (*content provider side*) → Content Delivery Network (CDN)



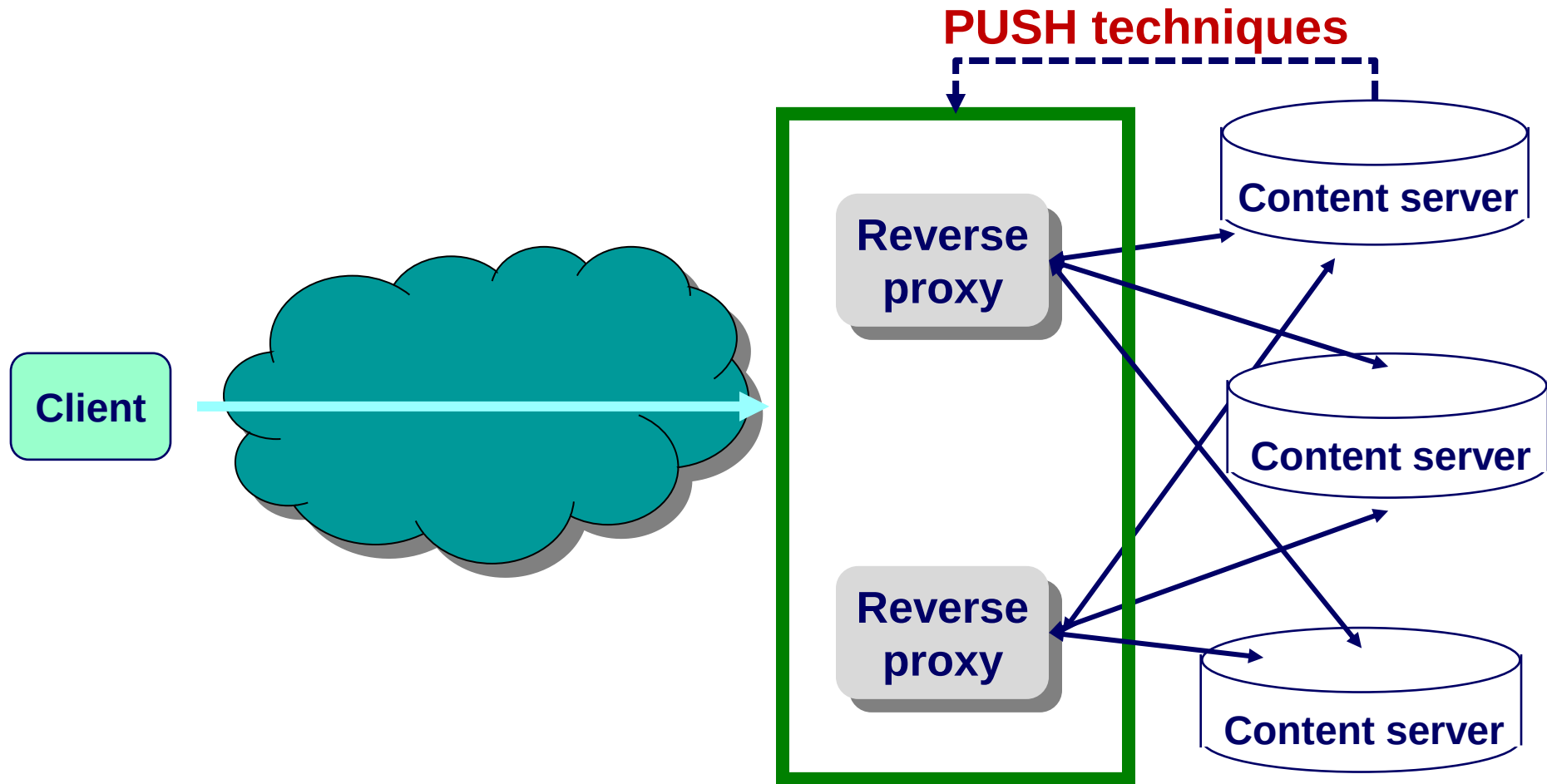
Proxy server: the origin of network cache servers

Initially: *Intranet gateway to Internet*

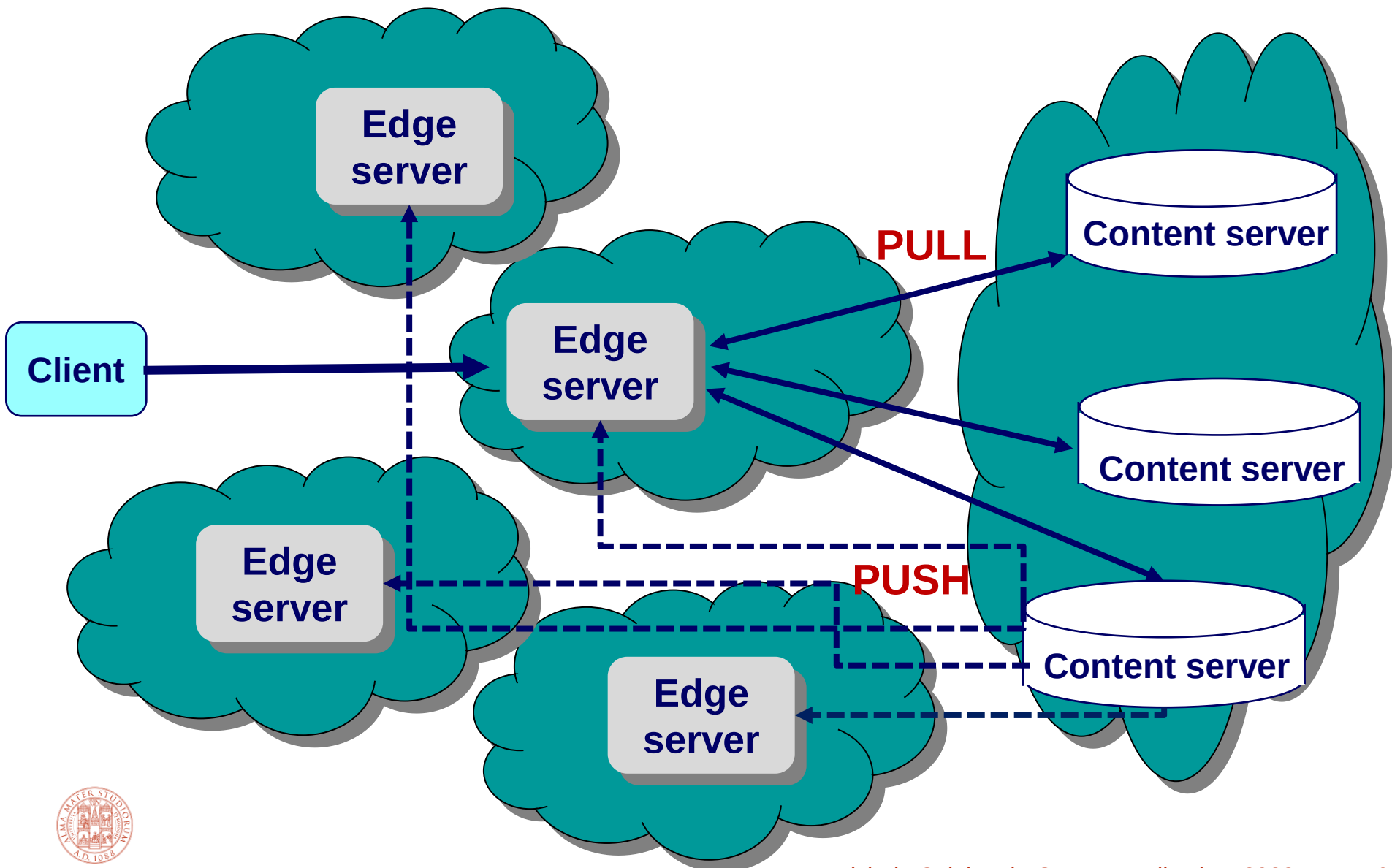
Then: *cache repository*



Reverse proxy (*CONTENT PROVIDER* side)



Possibility of spreading proxy servers in different network regions (AS)



Proxy and Reverse proxy advantages

(A) Proxy (B) Reverse (C) Both

- Reduce client ISP costs (A)
- Intelligent load sharing among origin servers (B)
- Reactively replicate and push popular content (C)
- Reduce origin server costs (C)
- Content reassembly and possibly adapt on behalf of the origin servers (C)
- Smaller round-trip times to clients (C)
- Maintain persistent connections to avoid TCP setup delay, e.g., handshake and slow start (C)



Limitations of Web caching

- **Much content is not cacheable**
 - Volatile data: stock prices, scores, web cams
 - Scripts: results depend on parameters
 - Cookies: results may depend on passed data
 - SSL: encrypted data is not cacheable
 - Analytics: owner wants to measure hits (and popularity)
- **Stale data**
 - Overhead of refreshing cached data

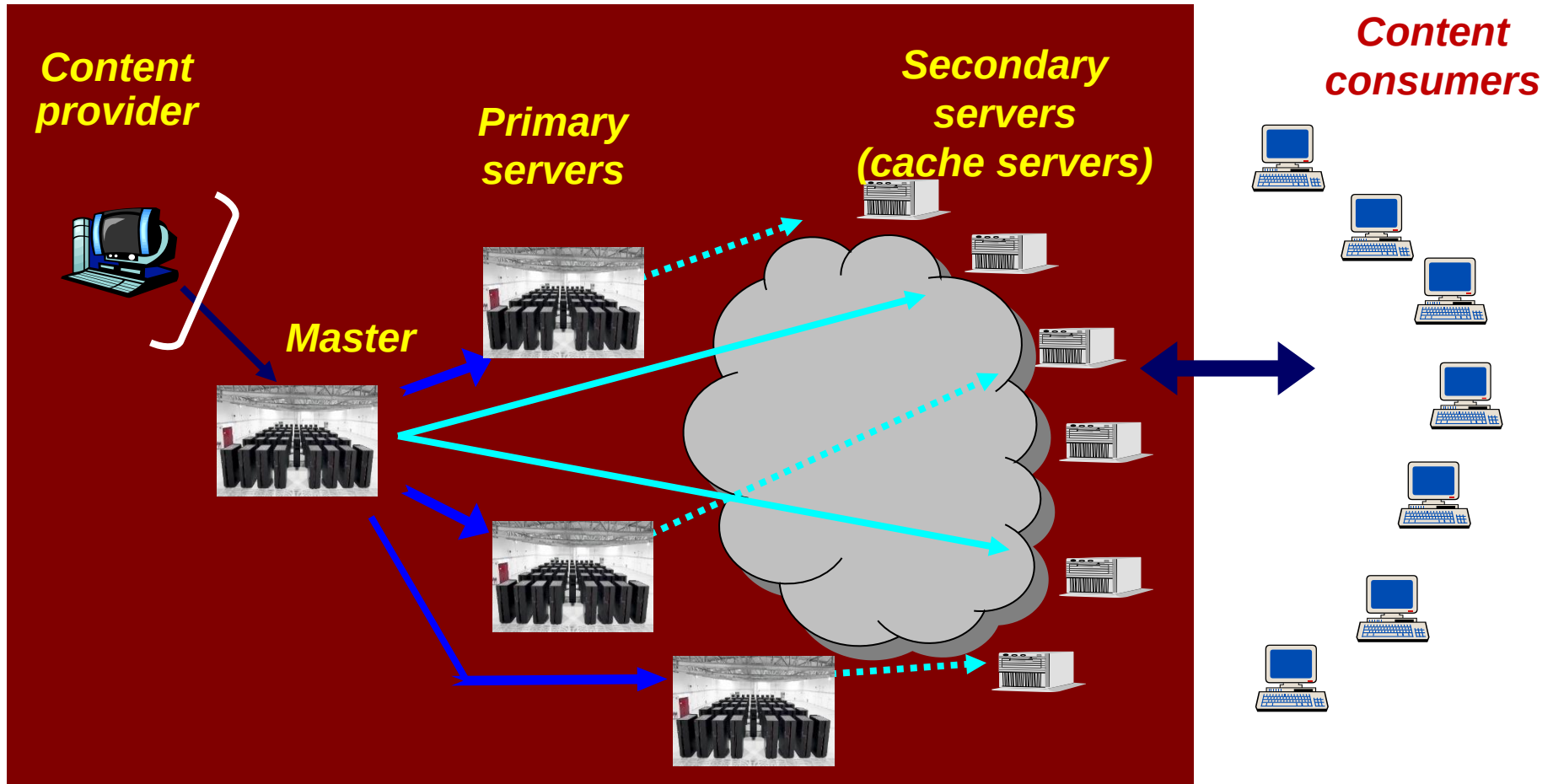


Management of large static content (e.g., video)

- Download “content list” from the origin server
- List of video segments belonging to video
 - Each segment is 1-2 seconds in length
 - Client can know time offset associated with each segment
 - Standard naming for different video resolutions and formats: e.g., 320dpi, 720dpi, 1040dpi, ...
- Client downloads video segment (at certain resolution) using standard HTTP request
 - Requests can be satisfied by cache because static objects
- Client observes download time vs. segment duration and might increase/decrease resolution if appropriate



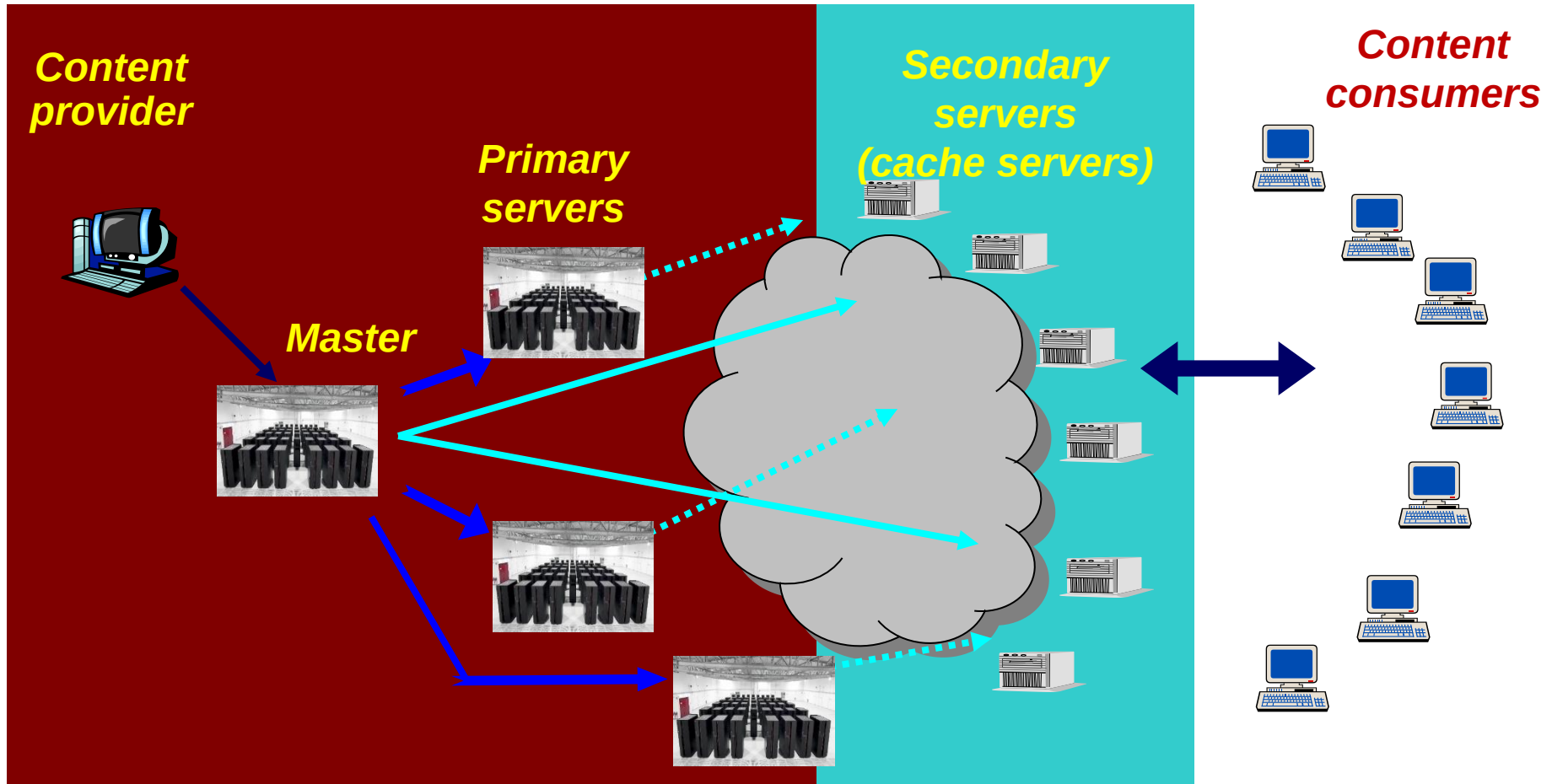
Content provider owns and manages everything



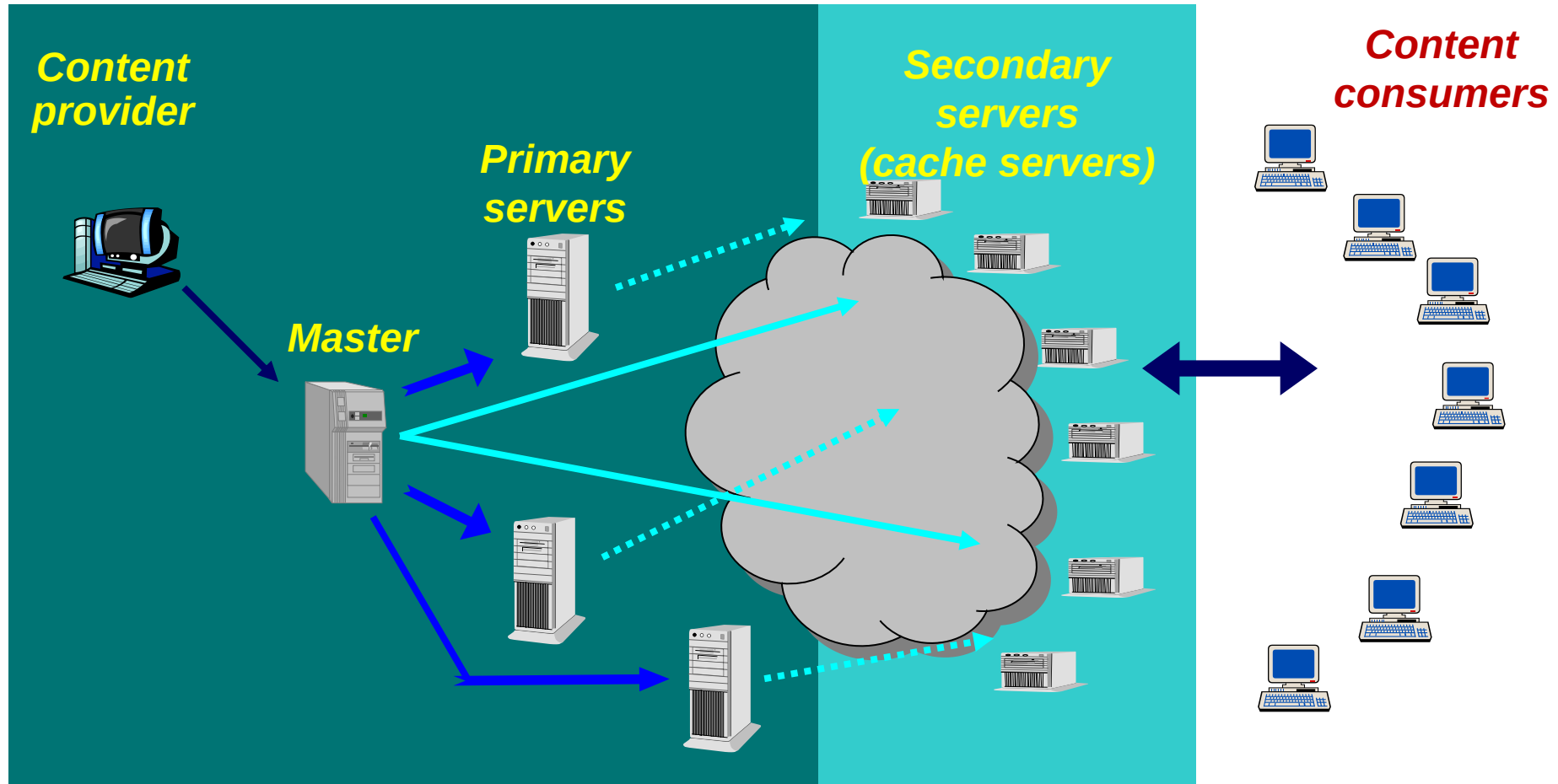
Example: “Office 365” sites



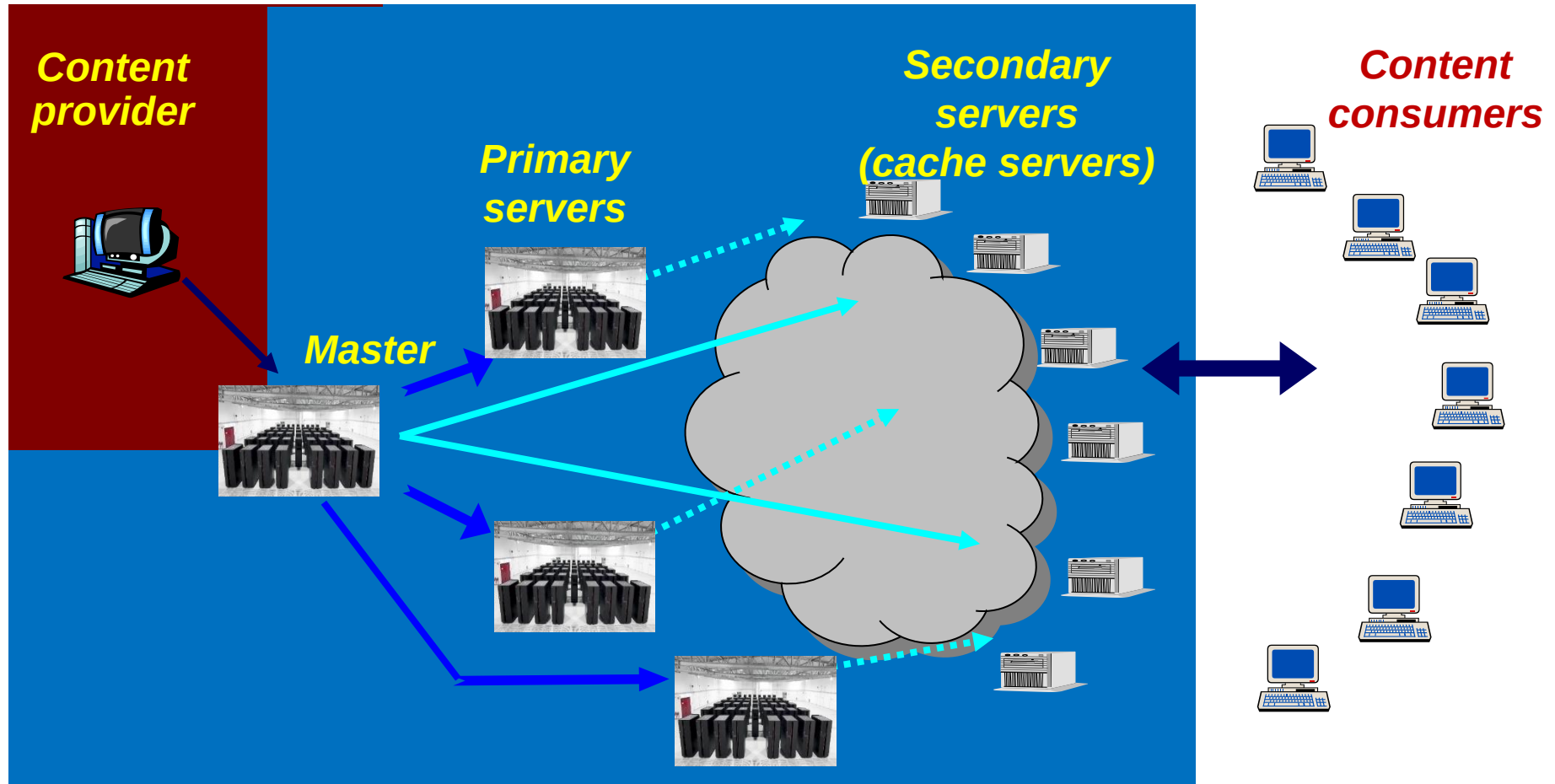
Content provider owns and manages its primary servers



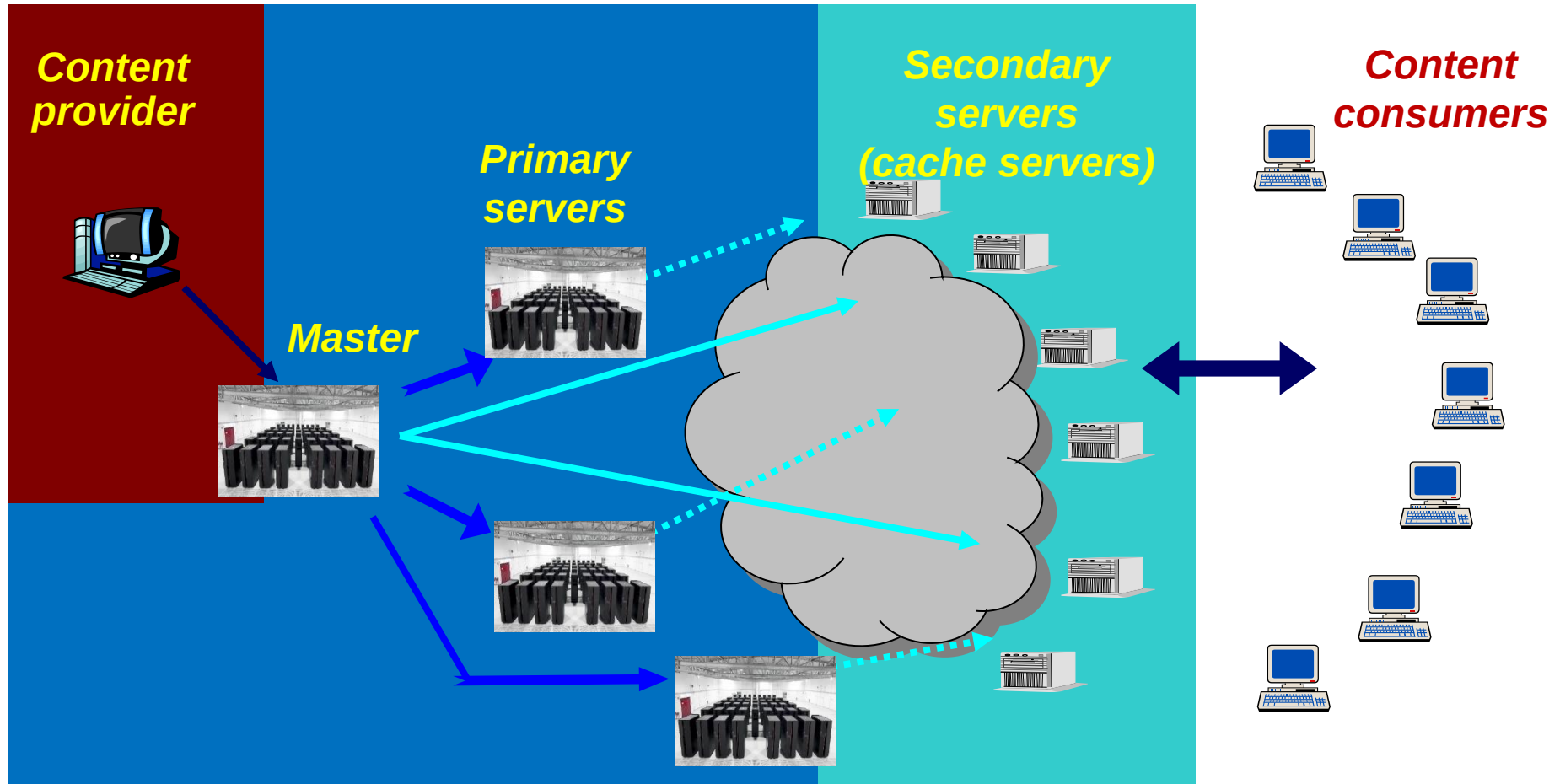
Content provider manages (virtual) primary servers hosted by another provider



Content provider generates just content and uses servers of one third party provider

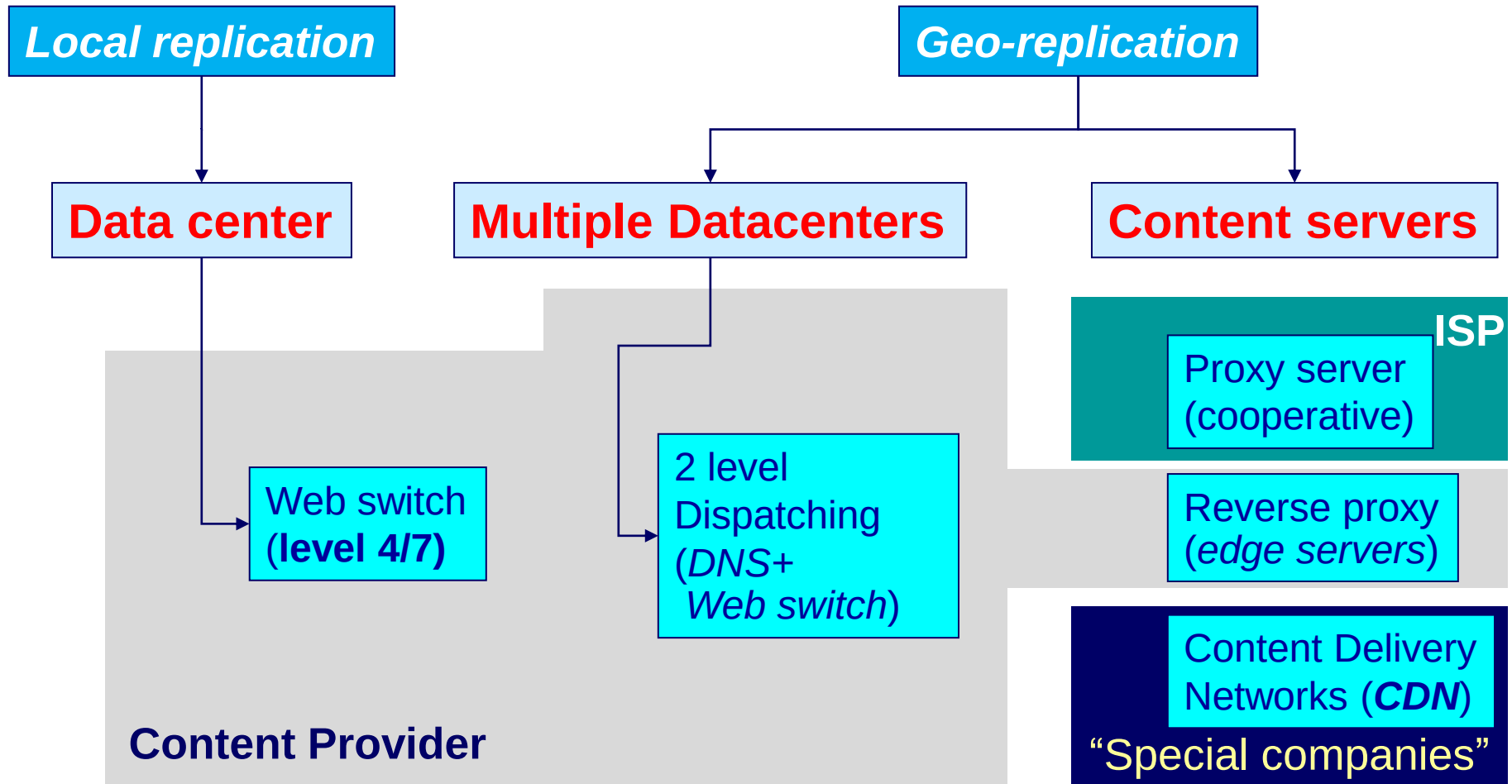


Content provider generates just content and uses different providers

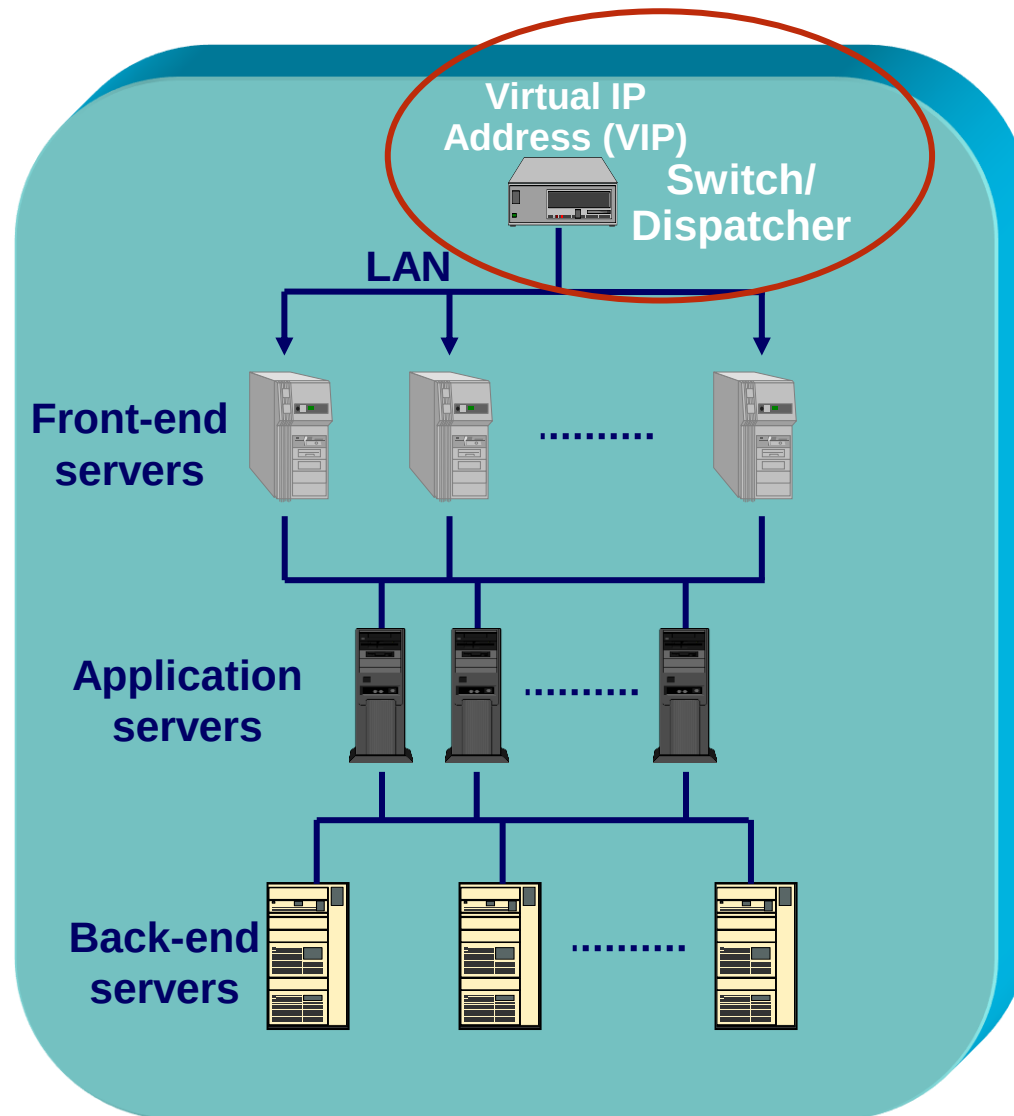


Server and Data replication

(Content provider owns and manages primary server)



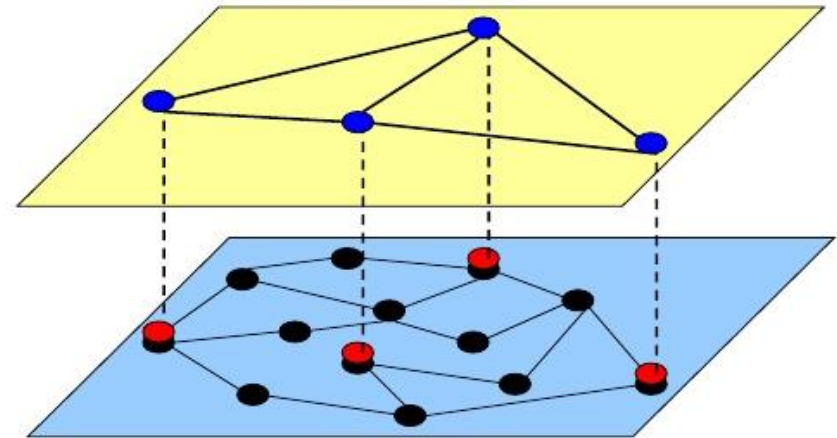
Local replication - Front-end switch



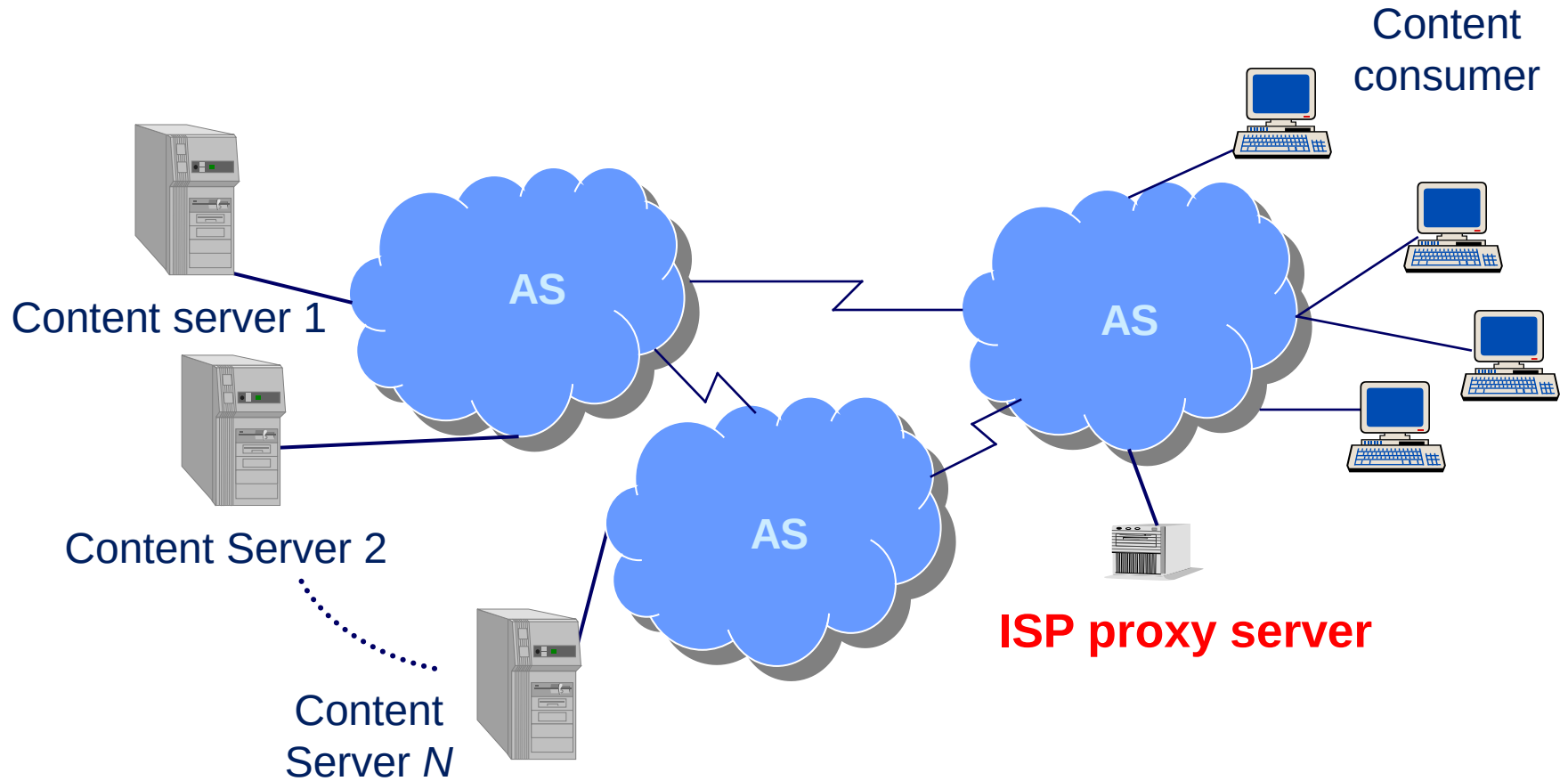
Geographical replication through *overlay network*

An overlay network is a “content-based” network built on top of one or more underlying networks

- Adds an additional layer of indirection (*virtual routing*)
- Changes properties in one or more areas of underlying network
- Integrate network routing with some computing functions (e.g., caching, adaptation, application-based routing)



Web caching - ISP

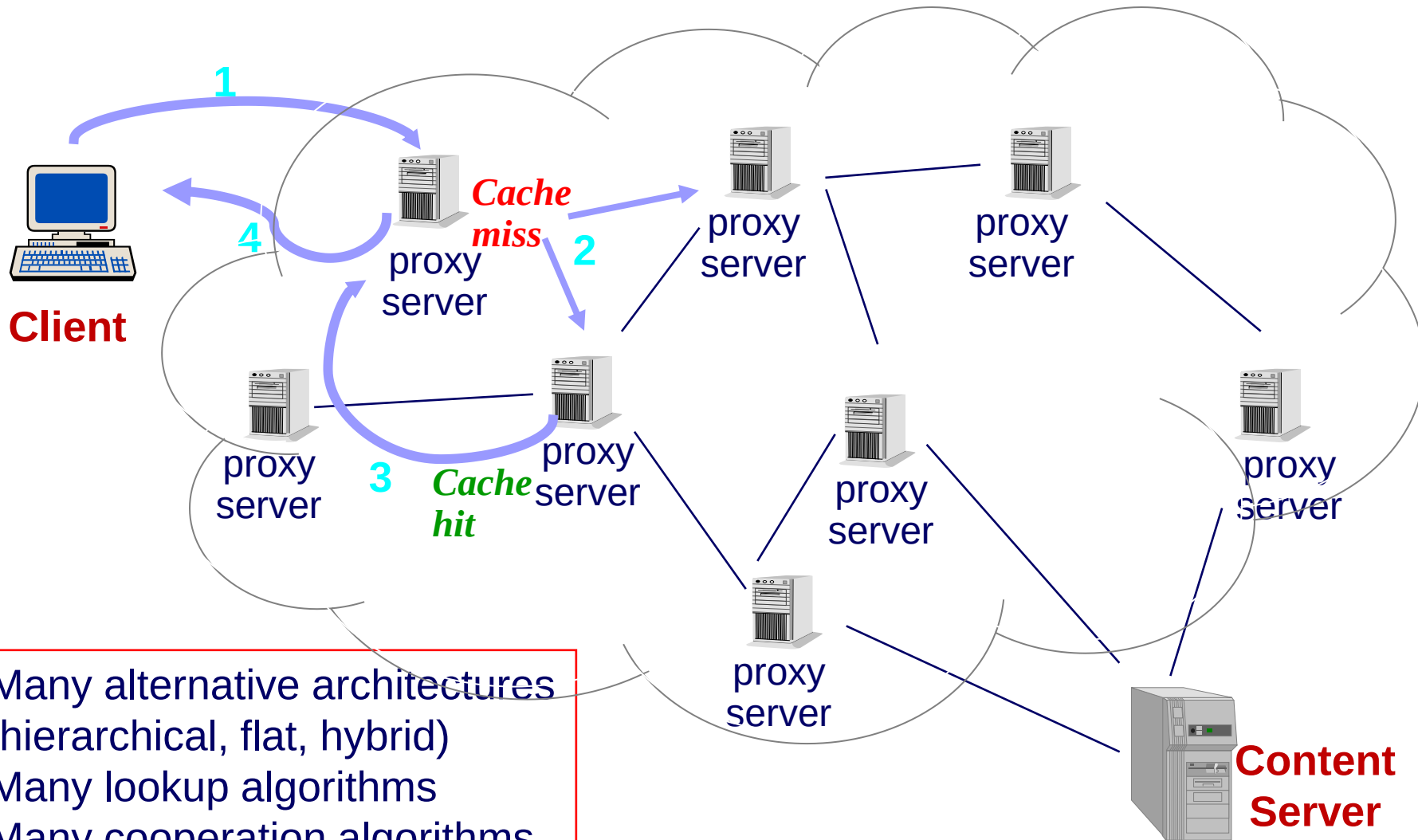


Cooperative network caching

1. **ISP** (Internet Service Provider) → Different cache servers in different POPs of the same AS
2. **Content Delivery Networks** → A cache server in each principal AS of the world



Cooperative caching (same ISP)

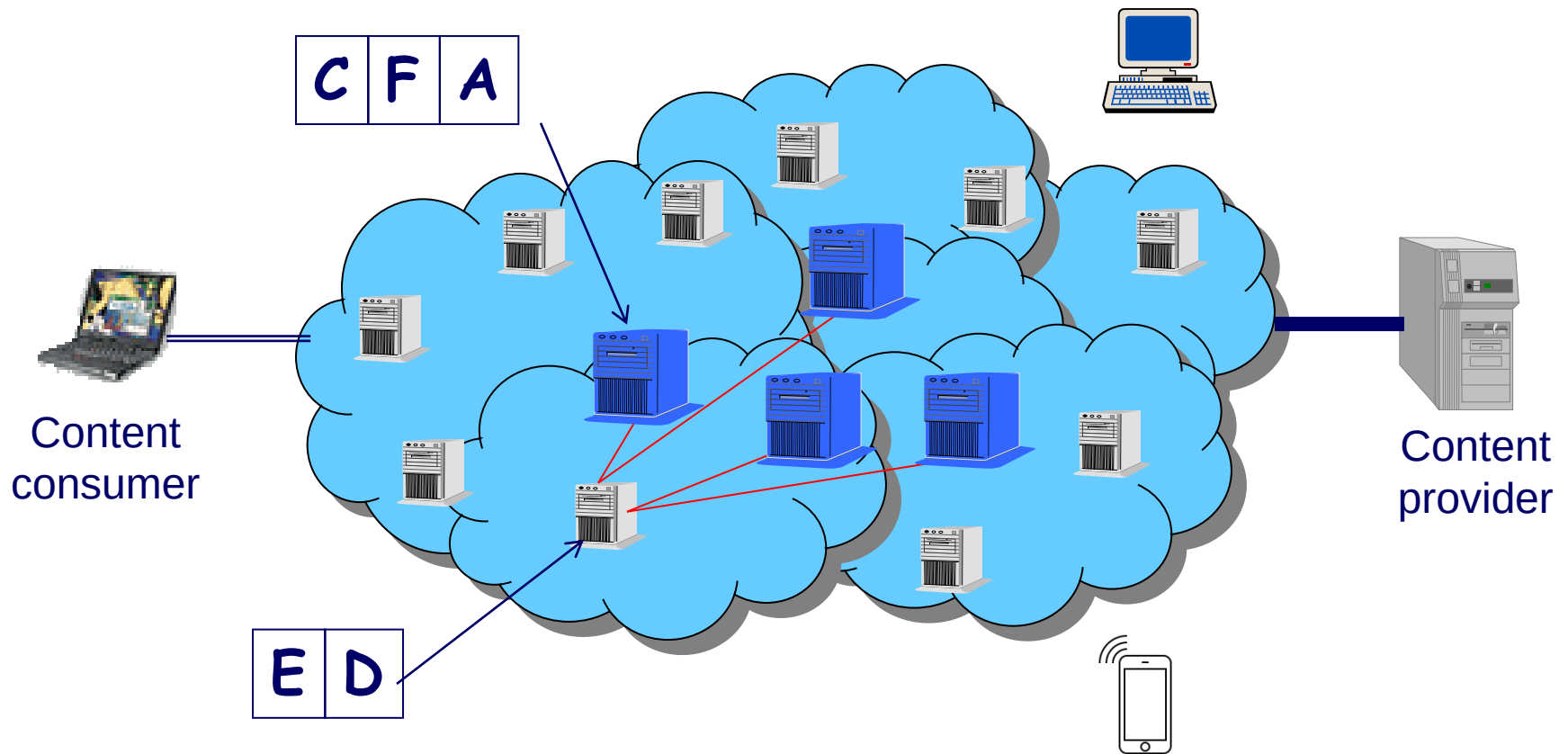


- Many alternative architectures (hierarchical, flat, hybrid)
- Many lookup algorithms
- Many cooperation algorithms



Messages:

- *Yesterday*: keep the edge servers as simplest as possible
- *Today*: assign computations at the edge (*fog computing*)



Pros and Cons of network caching

- *User benefits*: Latency reduction experienced by the user
- *Community benefits*: Less network traffic
- *Provider benefits*: Less overhead on datacenters

Unfortunately, generalized caching, even at cooperative level, has limited benefits (*low cache hit rates, not cacheable content*)



SELECTIVE CACHING



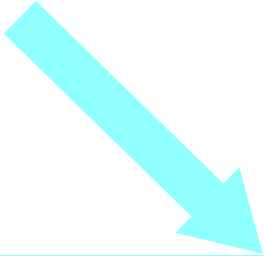
Motivation

“Content” delivery $\neq$ data delivery

Content = digital material for which “someone”
(*user, ISP, content provider, ...*) is ready to pay

Examples

- large embedded objects
- banners
- video-on-demand
- TV on demand
- *pay-per-download* music
- software distribution
- software *pay-per-use*



Services must be characterized by:
Top performance and reliability
(*best effort is not acceptable*)

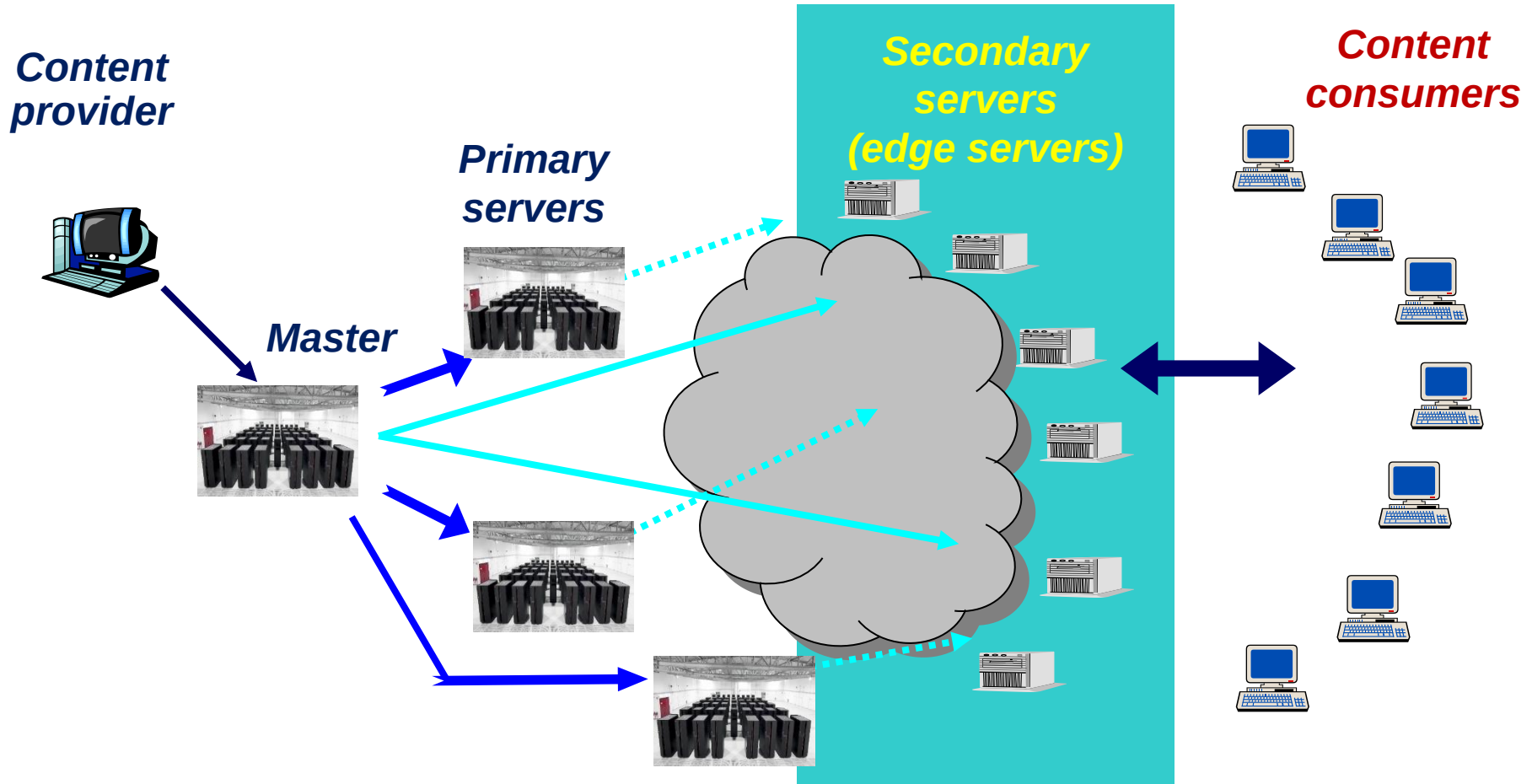


Content Delivery Networks (CDN)

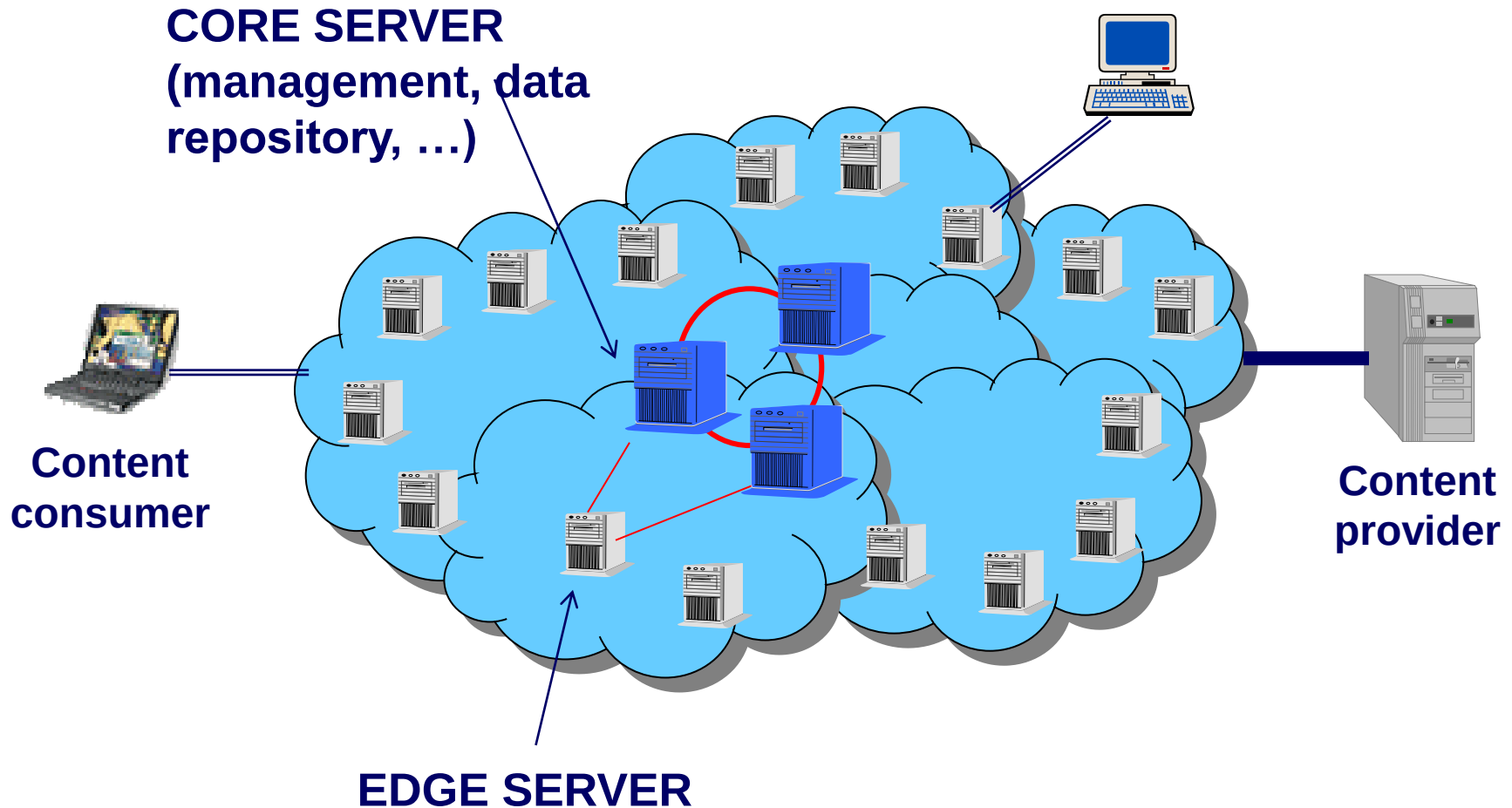
- CDNs provide a service of *geographically distributed cooperative caching for selective content* (that is, just for the services of their customers)
- They can achieve high *cache hit rates* thanks to:
 - **Selective caching**
 - **Pro-activity** (*pushing* and not only *pulling* of content)
 - **Data synchronization** among content replicas



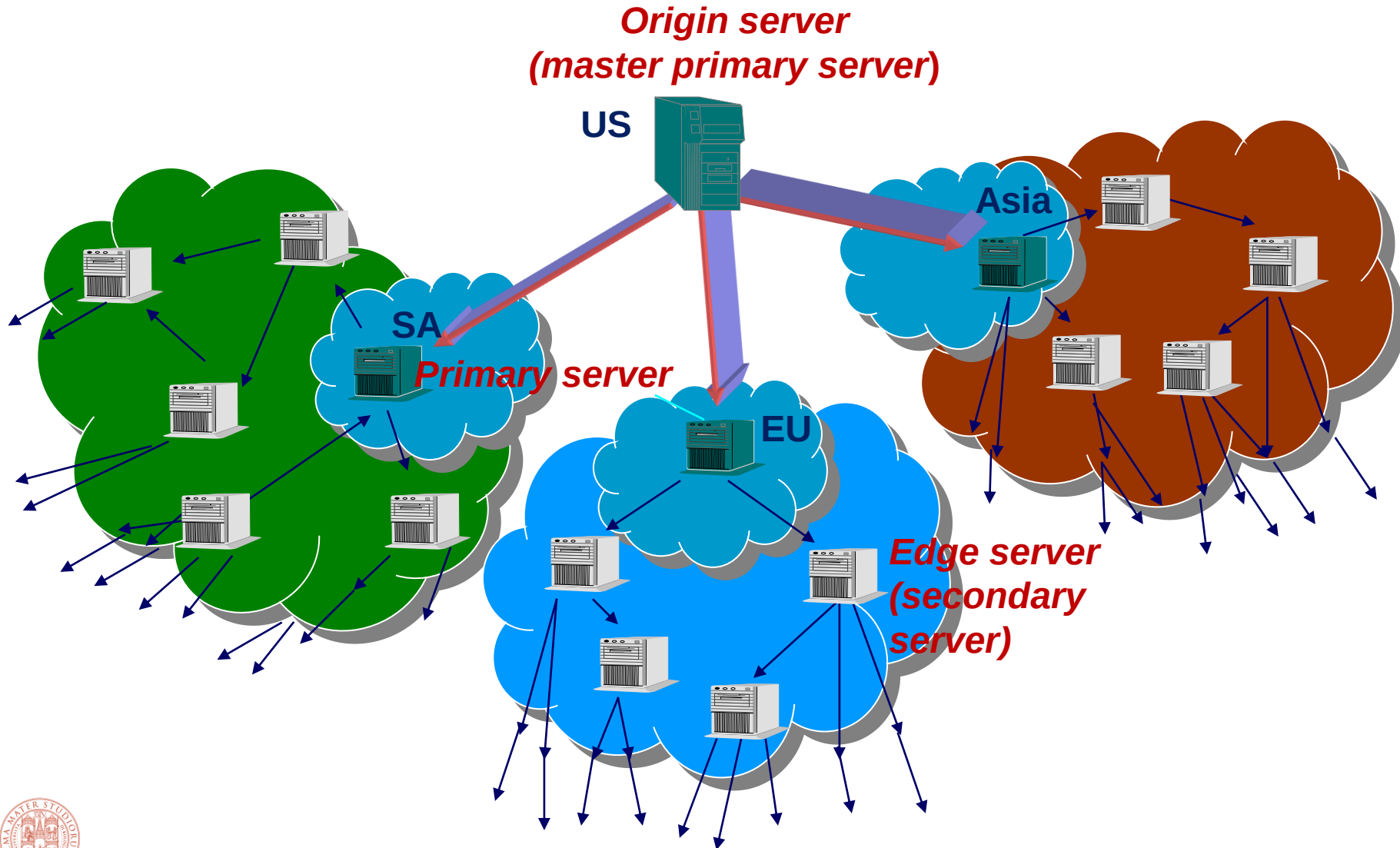
Content provider owns and manages its primary servers



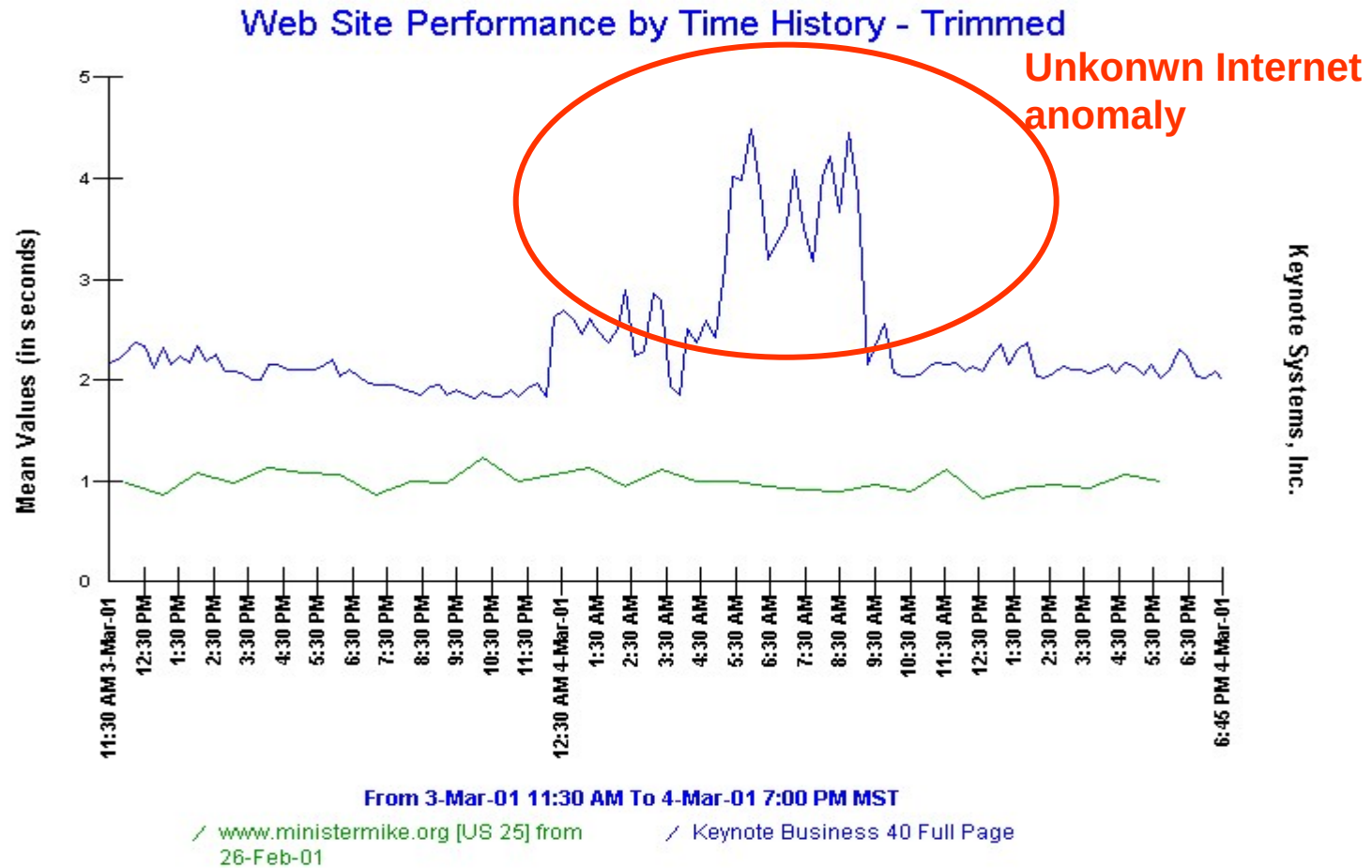
CDN architecture: Two main server levels



The overall picture



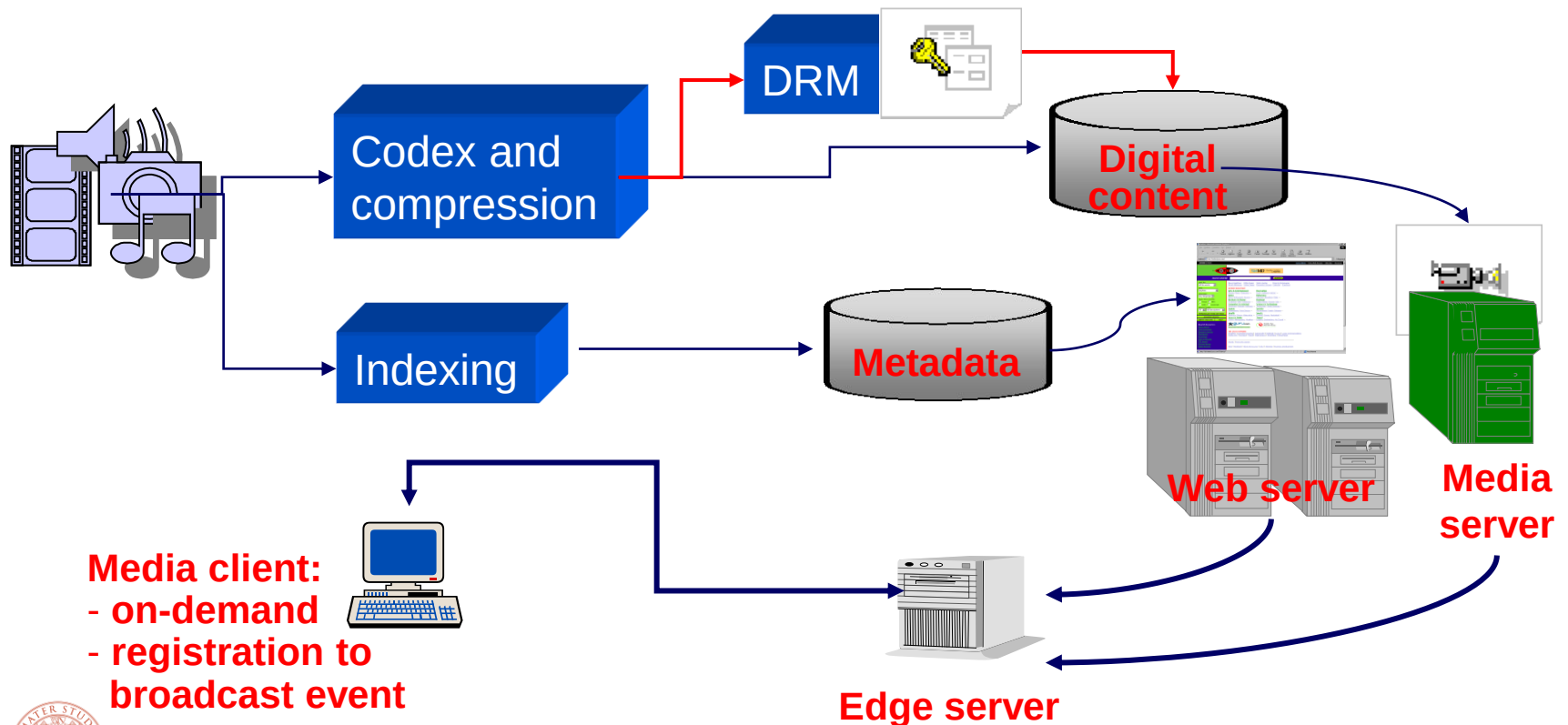
Does CDN work? Yes



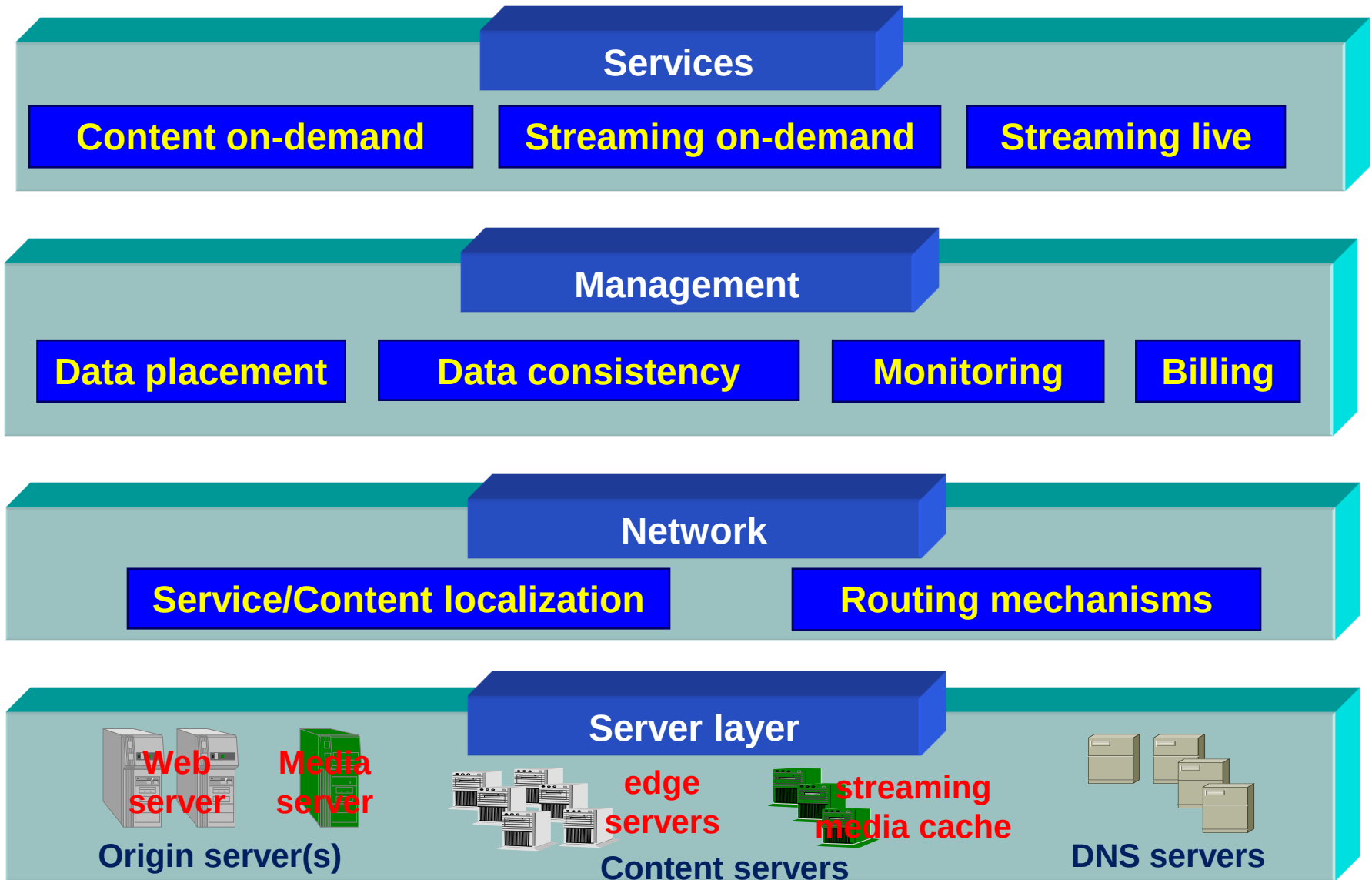
Multimedia delivery

The architecture consists of two main components:

- *encoded media*
- *metadata*



CDN components

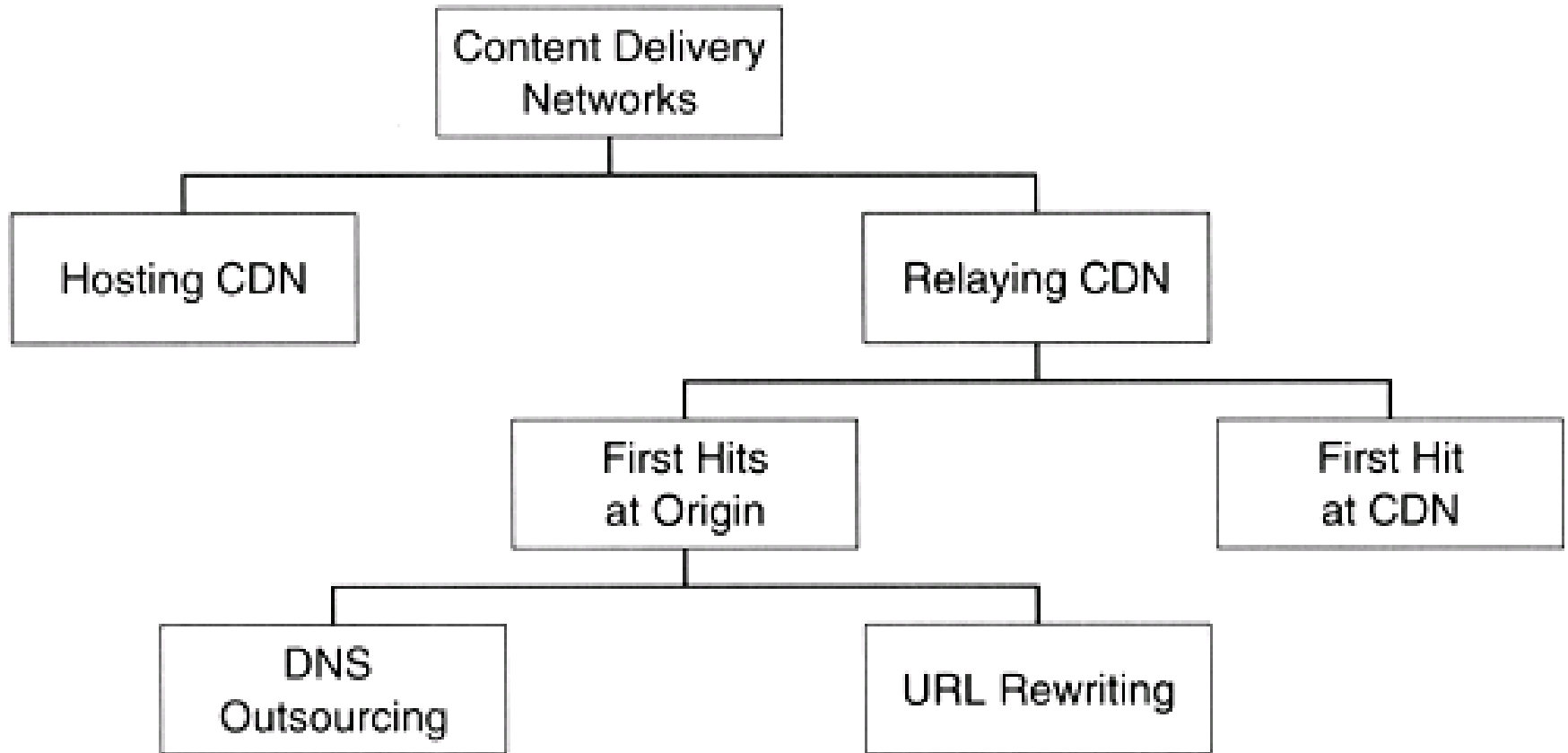


Main CDN functions

- ***Request redirection and content delivery services*** to direct a request to the closest suitable edge server using “intelligent” mechanisms to bypass congestion
- ***Content outsourcing and distribution services*** to replicate and/or cache content to distributed edge servers on behalf of the origin server
- ***Content negotiation services*** to meet specific needs of each customer → ***Service Level Agreements***
- ***Management services*** to manage the network components, to handle accounting, and to monitor and report on content usage

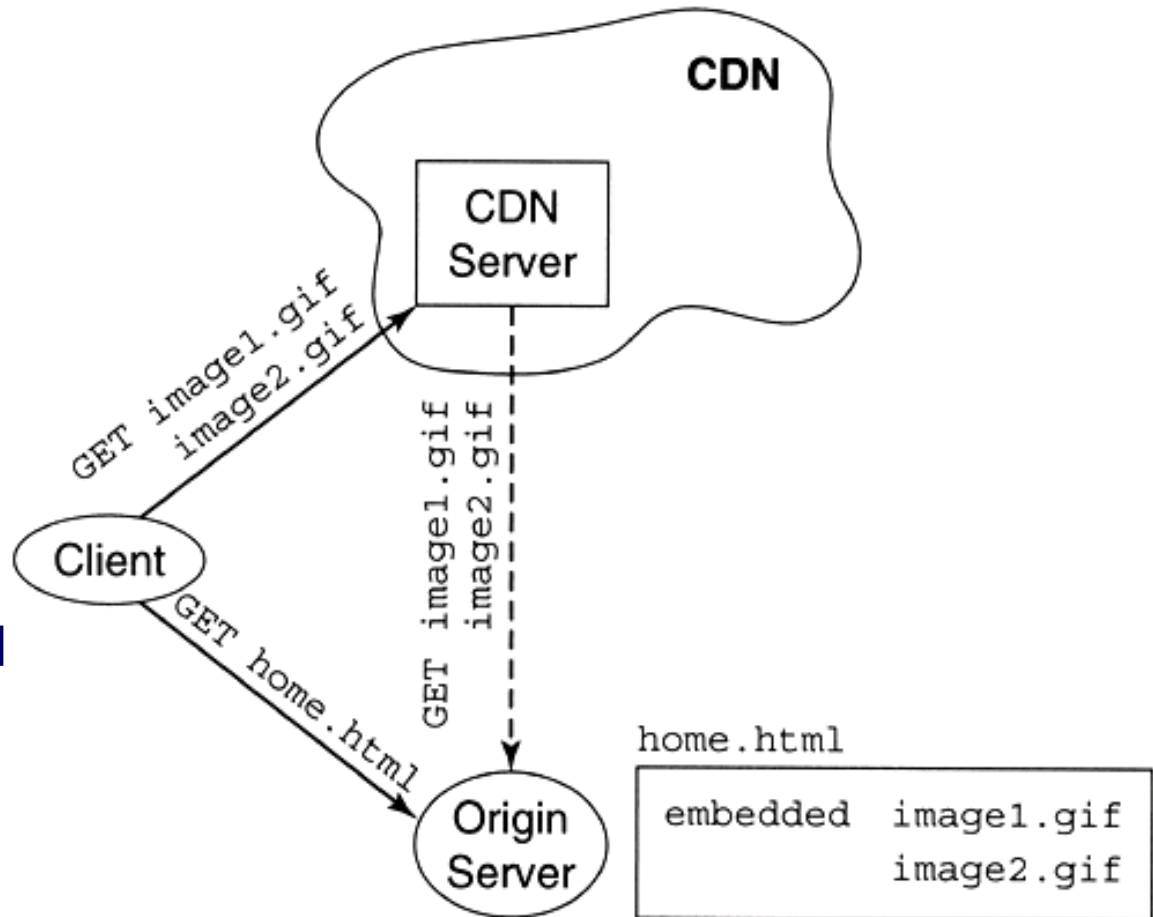


CDN types

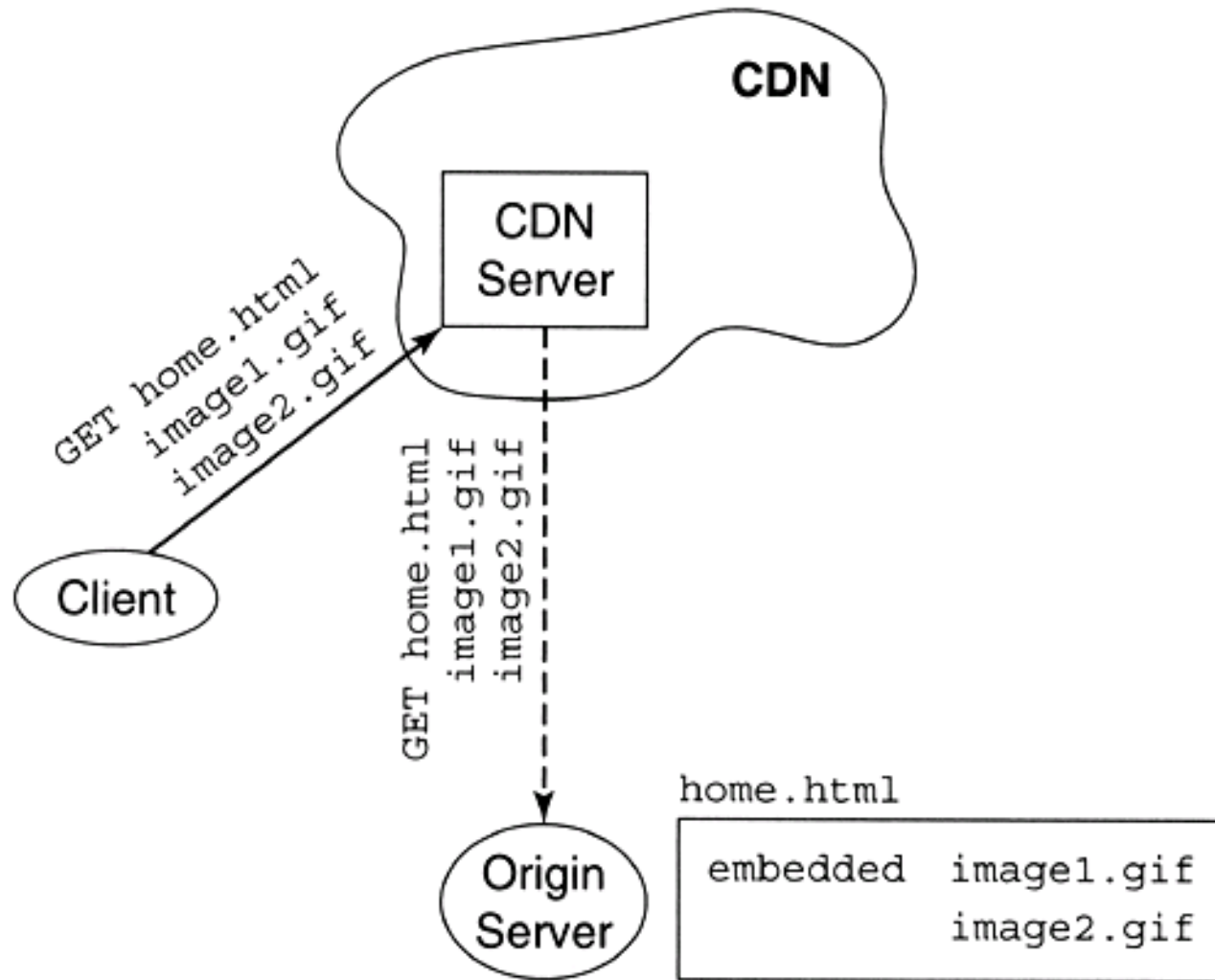


Relaying CDN: 1st hit at the origin server

- Embedded URLs that the CDN is supposed to deliver use host names that are controlled by the DNS of the CDN
- Alternatively, the embedded content uses domain names that belong to the CDN

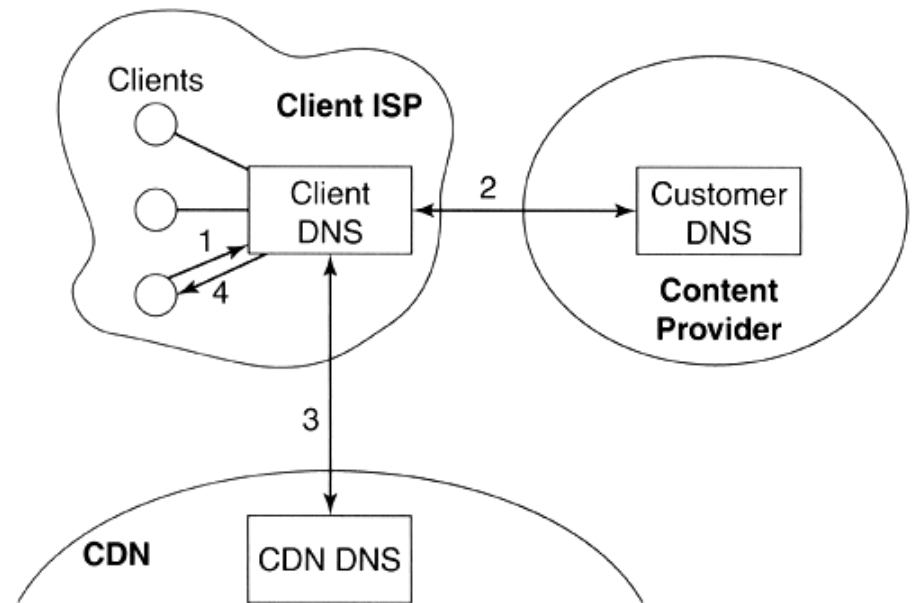


Relaying CDN: 1st hit at the CDN



Request delivery mechanisms

- DNS outsourcing
- Customer delegates the DNS service for its domain to the CDN
- Customer's DNS replies (when queried) with a point to the DNS of the CDN
- The DNS of the CDN resolves query to one of its CDN servers



Example: Akamai



CDN market leader

- Akamai deliveries almost 25% of Web traffic (at 6Tbps)
- It owns and manages
 - Multiple datacenters
 - Thousands of servers distributed in all main Autonomous Systems
 - Network segments
- Any type of customer interested to “content” delivery
 - TV broadcasting: CBS sport, ESPN, Fox sport, Adobe TV, ...
 - Radio broadcasting: National Public Radio, XM Satellite Radio, ...
 - Sport events: NBA, FIFA 2006 World Cup
 - Commercial events: Saatchi&Saatchi, Anheuser-Busch, Victoria's Secret
 - Social sites: Friendster, MSN Live Earth, MySpace, ...
 - Music and concerts: iTunes, Universal, MTV Networks, Warner Music, ...
 - Games: Sony Playstation 3, Nintendo Wii
 - Software companies for periodic patches: Microsoft, Adobe, ...



Some Akamai clients (>5000!)

- **TV Broadcasting**
 - CBS sport, ESPN, Fox sport, France TV Interactive, Adobe TV, ...
- **Radio Broadcasting**
 - National Public Radio, XM Satellite Radio, ...
- **Sport**
 - NBA, FIFA 2006 World Cup
- **Publicity**
 - Saatchi&Saatchi, Anheuser-Busch, Victoria Secret
- **Social network**
 - Friendster, MSN Live Earth, MySpace
- **Music and Concerts**
 - iTunes, Universal, MTV Networks, Warner Music Group
- **Gaming**
 - Sony Playstation, Nintendo Wii



How is it possible to intercept and route user requests to an edge server?

- **Combination of**
 - **URL modification** (*URL rewrite method*)
 - **DNS mechanisms**
- When a user clicks on a URL (www.xyz.com), if not cached, this request:
 - reaches the Authoritative name server of the “xyz” site
 - the Authoritative name servers of the companies using a CDN are managed (by contract) by the CDN
 - hence the AS-CDN can return the IP address of the edge server that is “**most convenient**” for that user, that content in that moment
- The “**most convenient**” goal opens interesting real-time issues:
 - **Systems for continuous passive and active monitoring (at network, server and content level), Data analytics, Internet proximity, Decision algorithms, etc.**

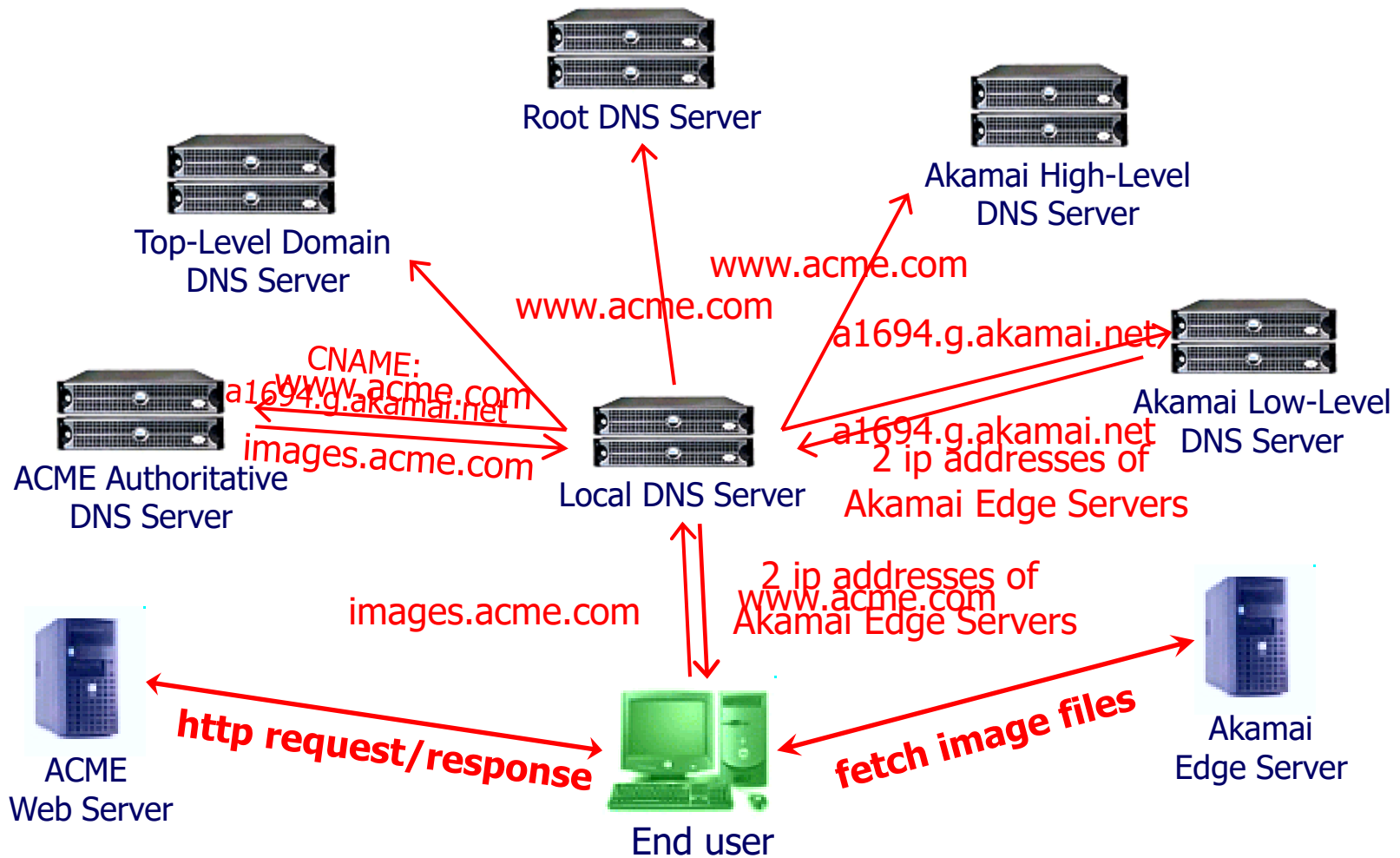


Akamai numbers

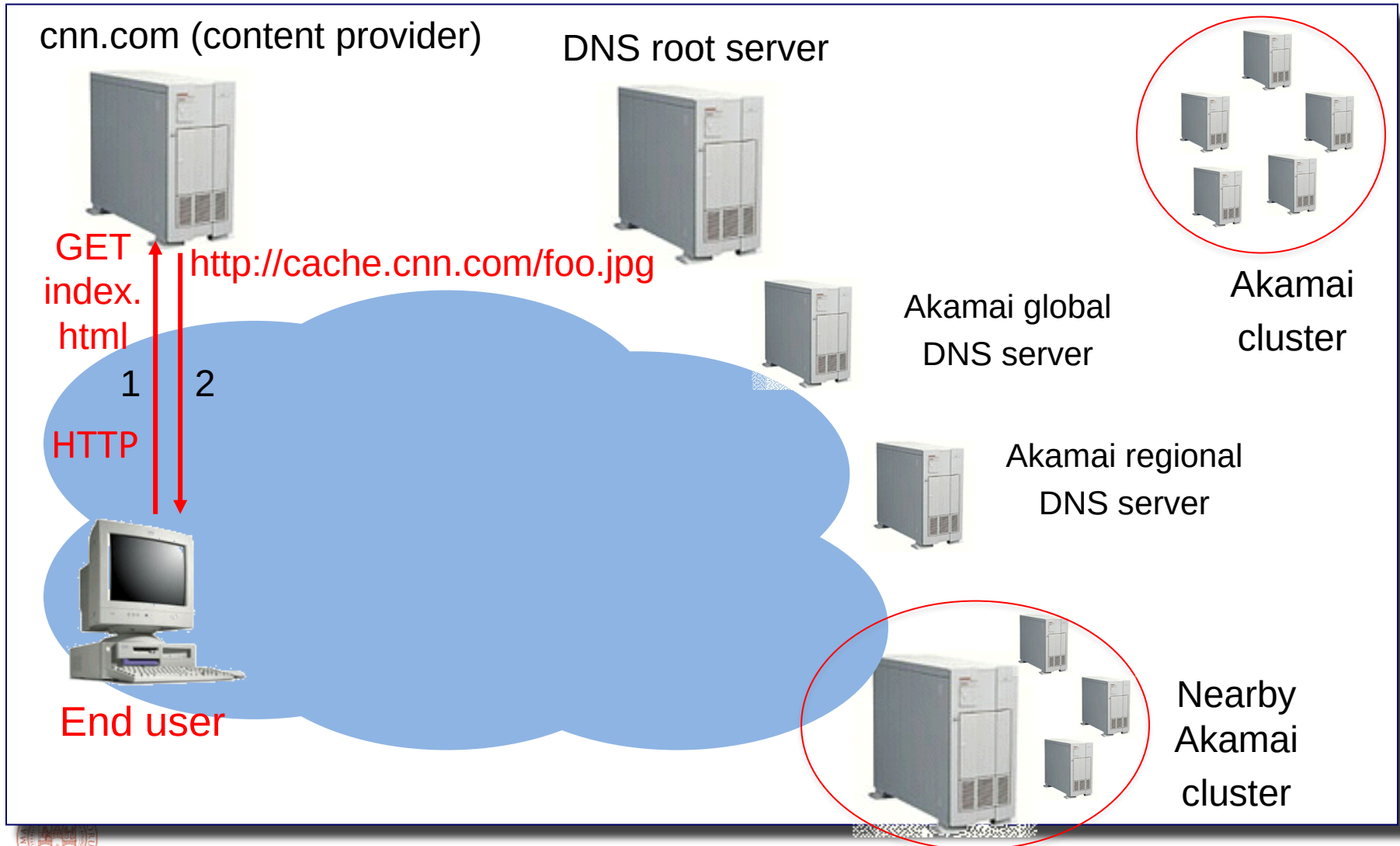
- **Distributed servers**
 - Servers: ~250,000
 - Networks: ~2,000
 - Countries: ~90
- **Client requests**
 - Hundreds of billions per day
 - Half in the top 45 networks
 - 20-25% of all Web traffic worldwide



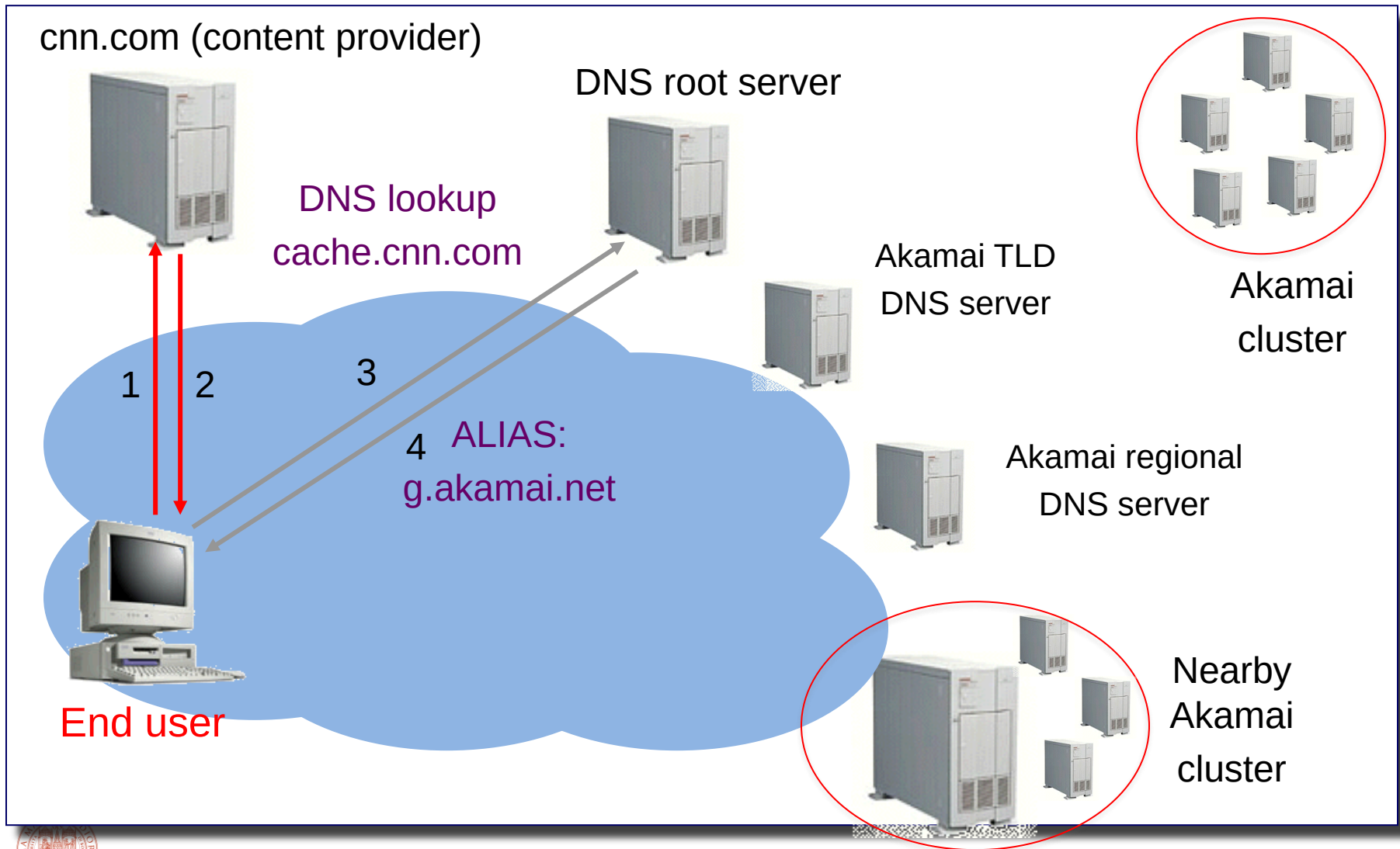
The overall picture (*not hosted Web content*)



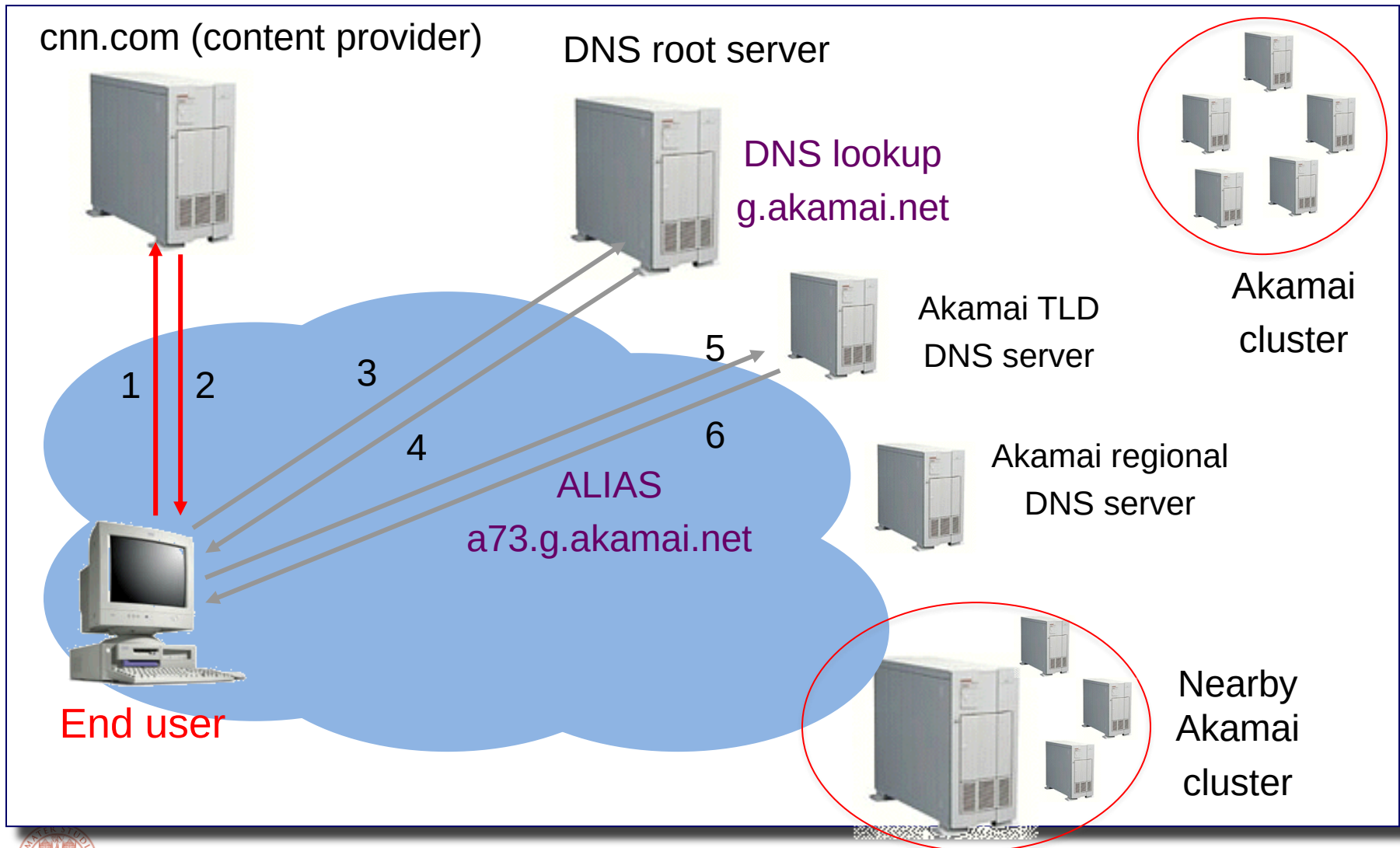
How Akamai uses DNS



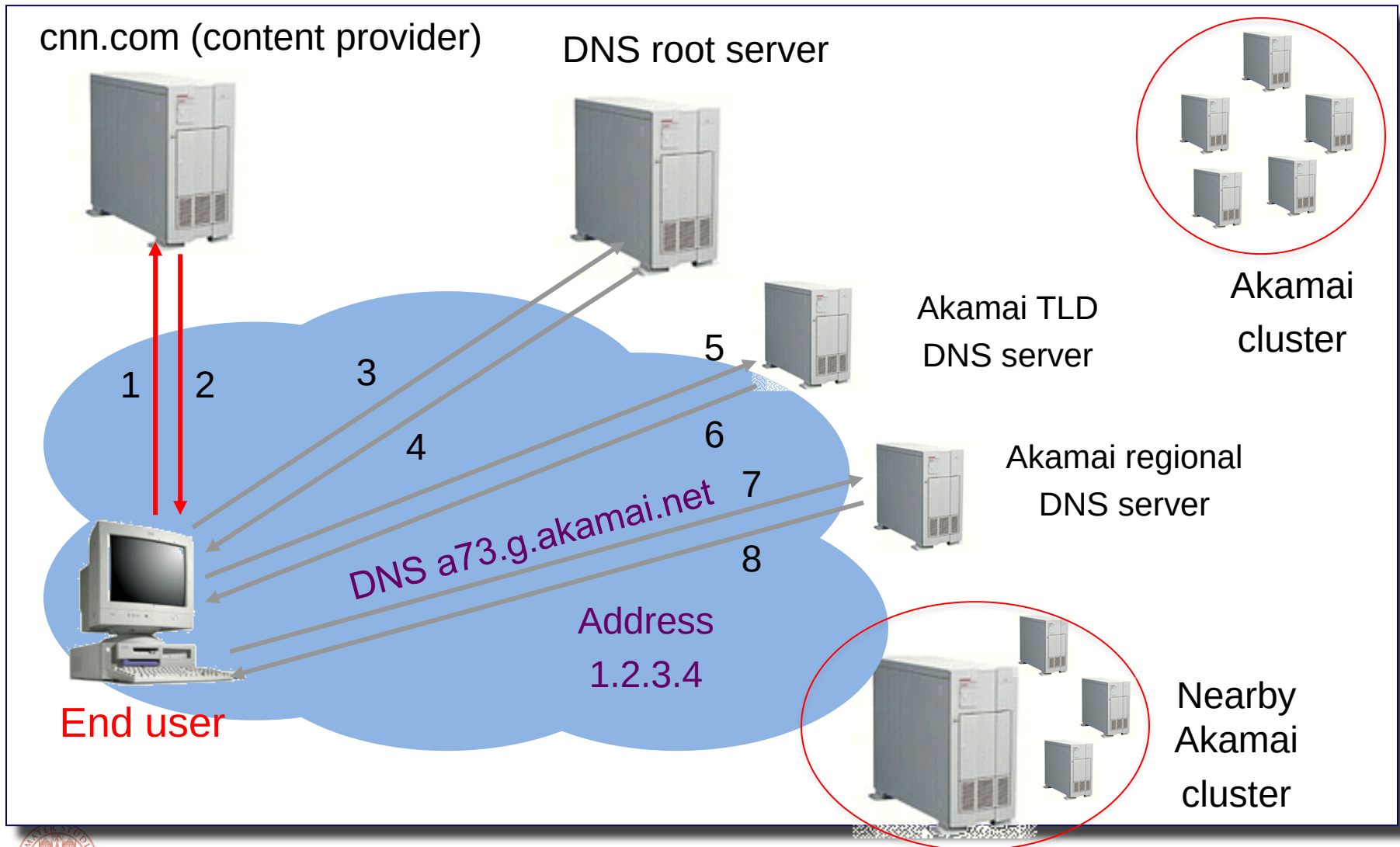
How Akamai uses DNS



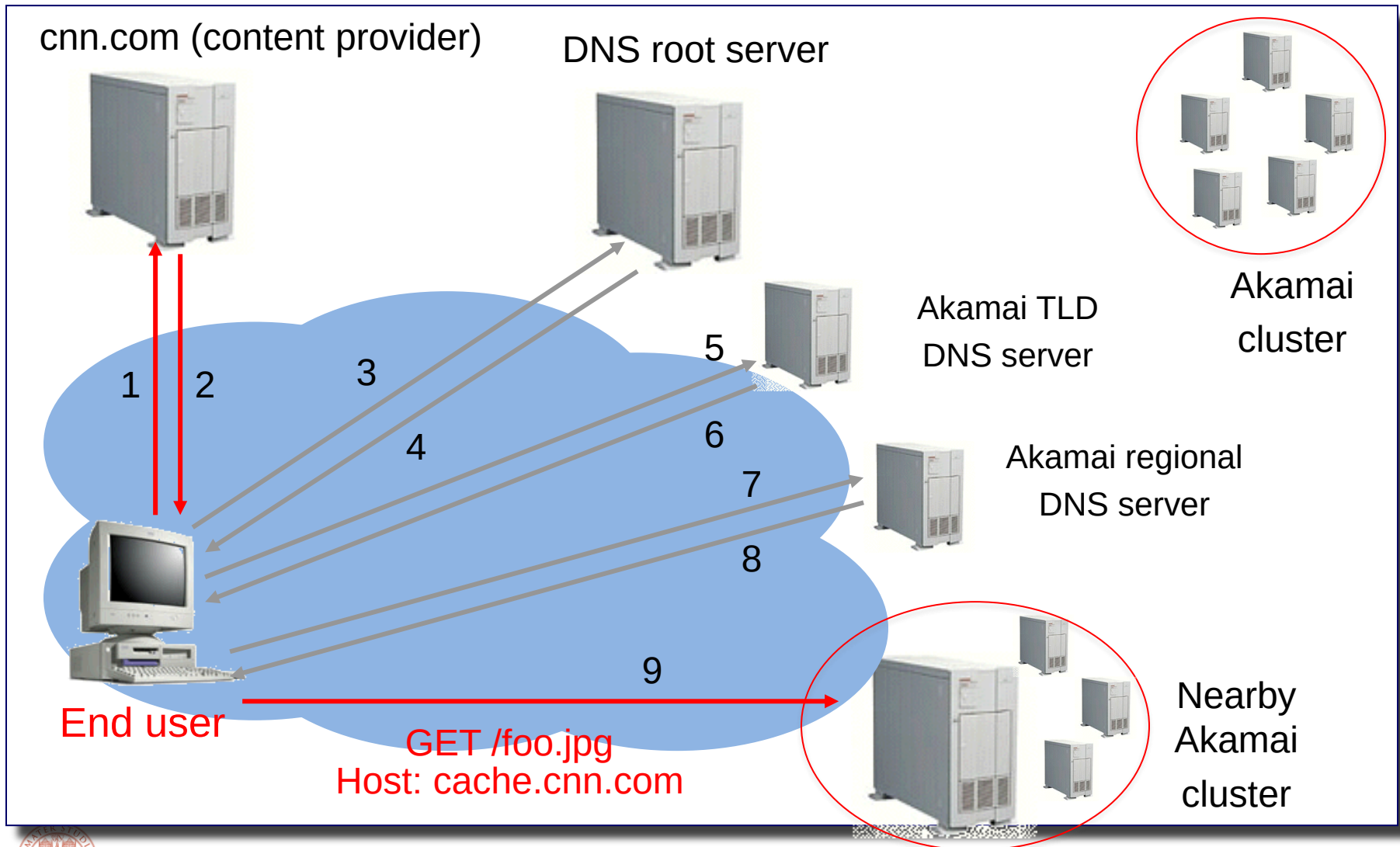
How Akamai uses DNS



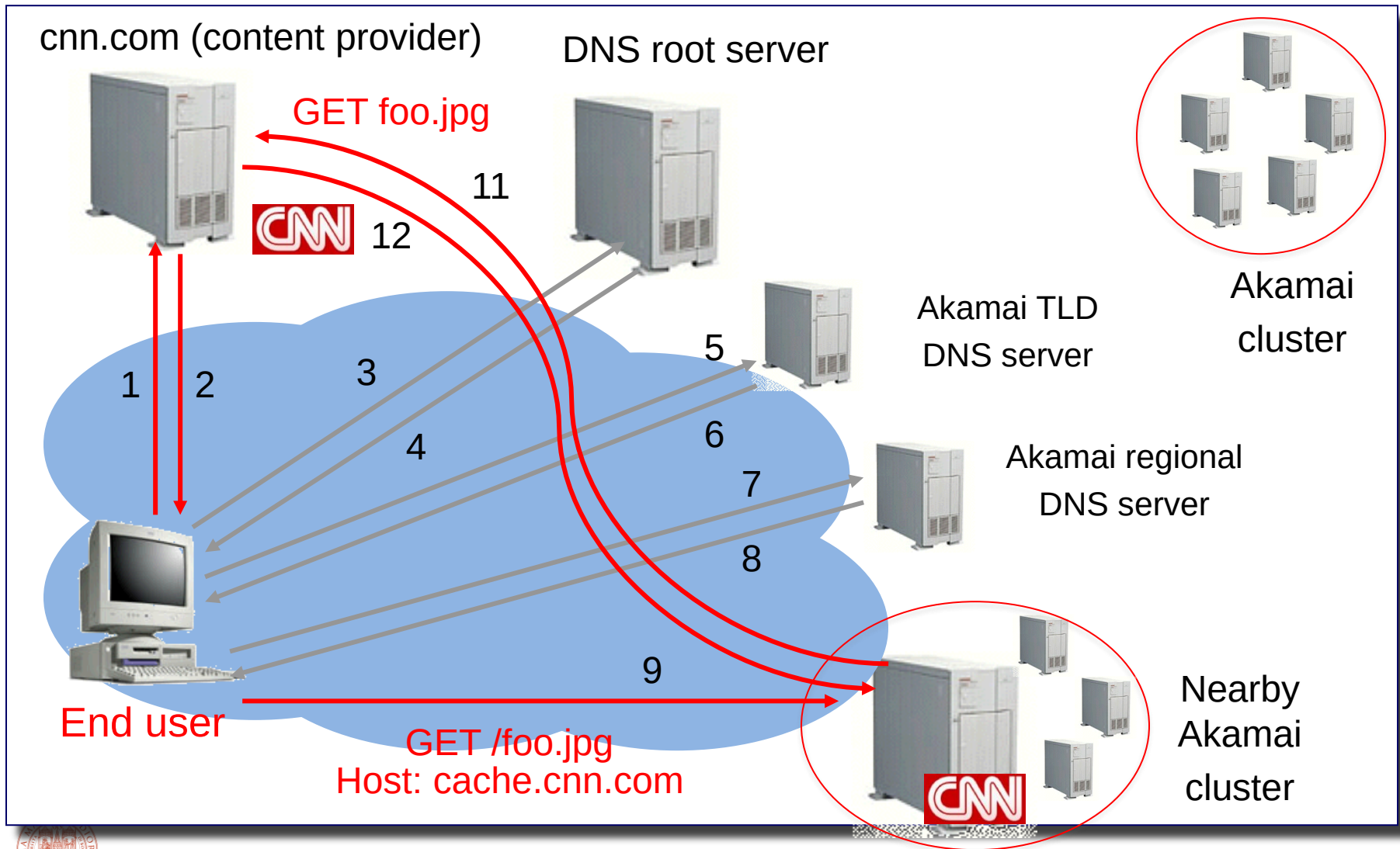
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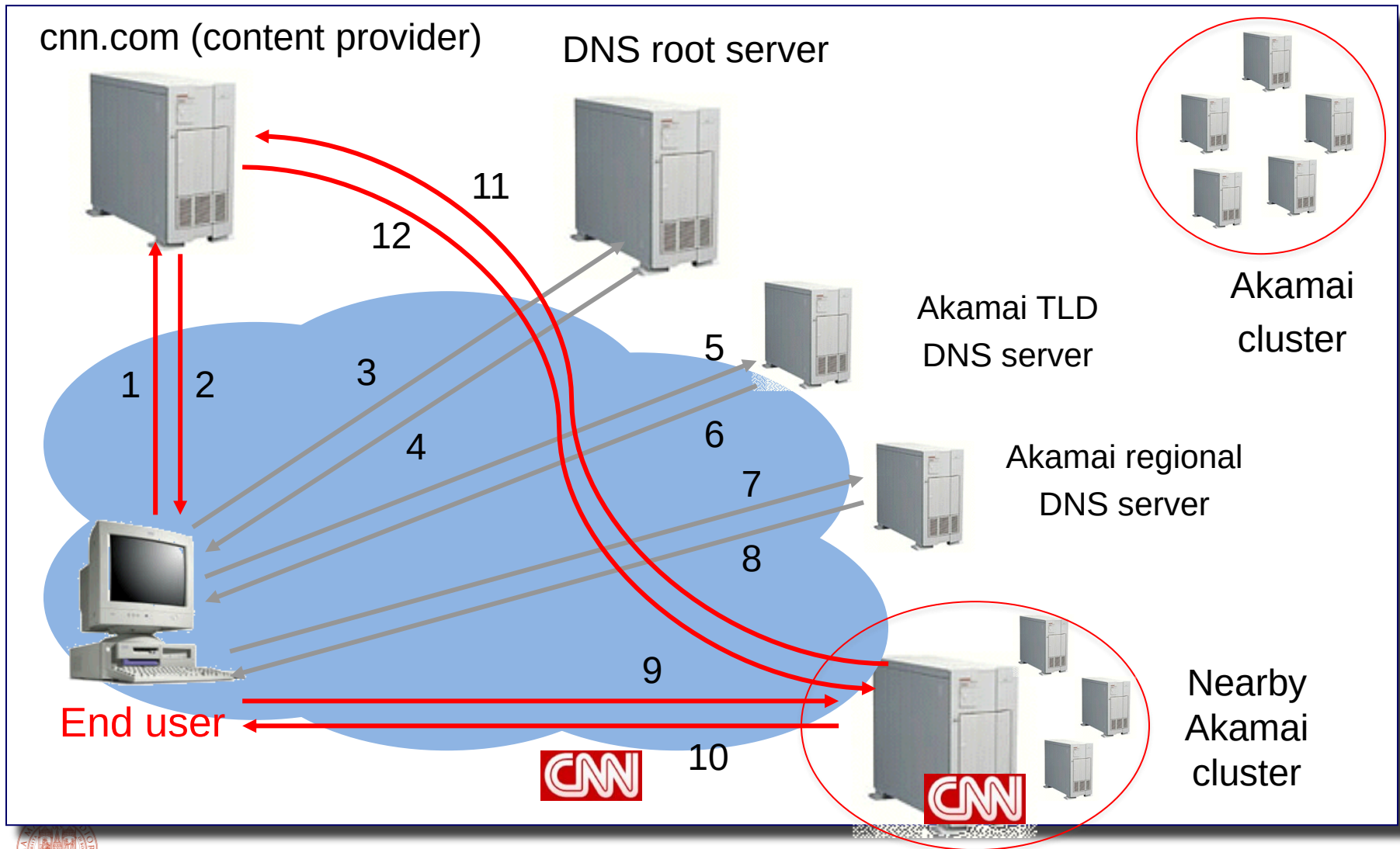
How Akamai uses DNS



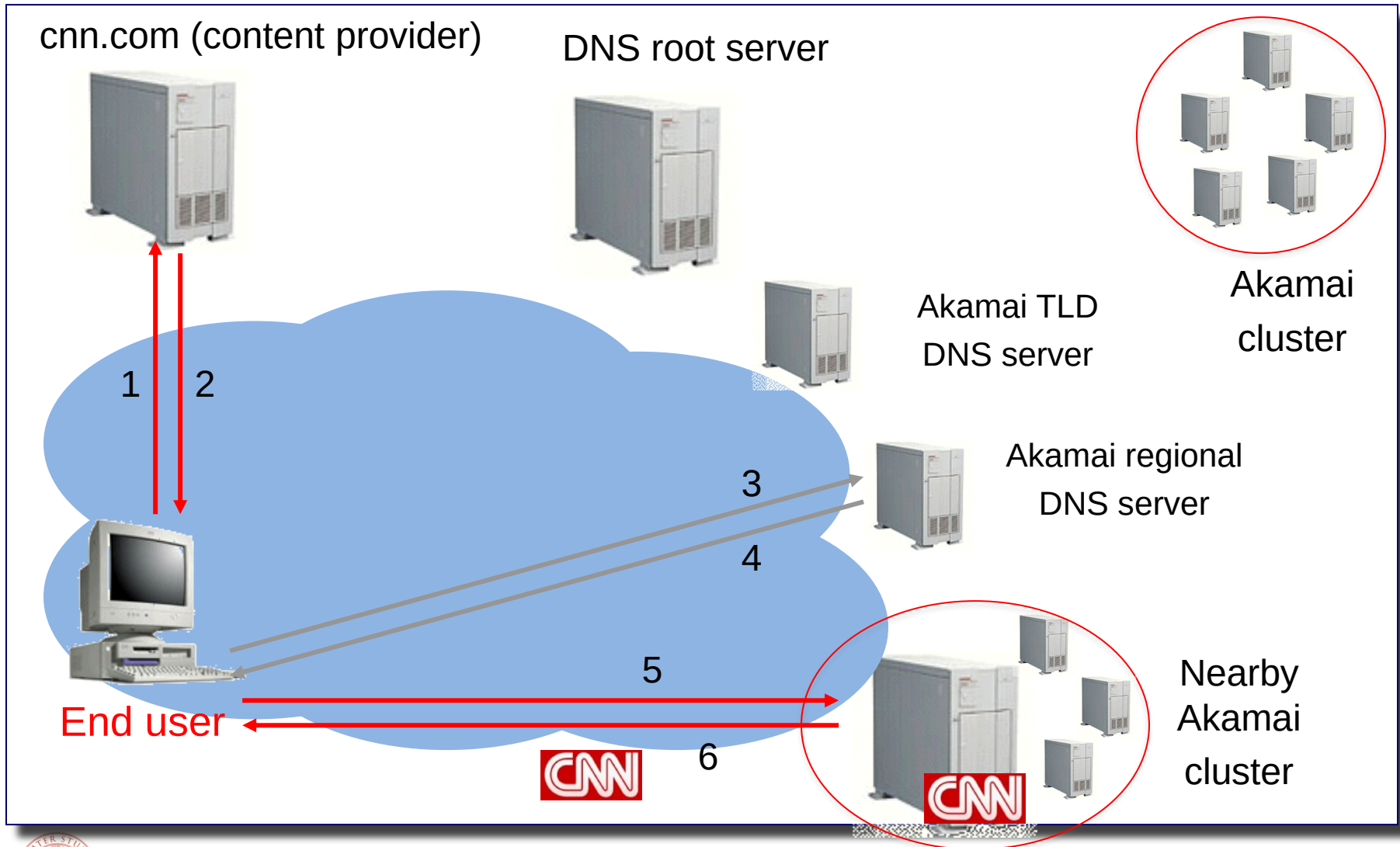
How Akamai uses DNS: *Cache miss*



How Akamai uses DNS: *Cache miss*

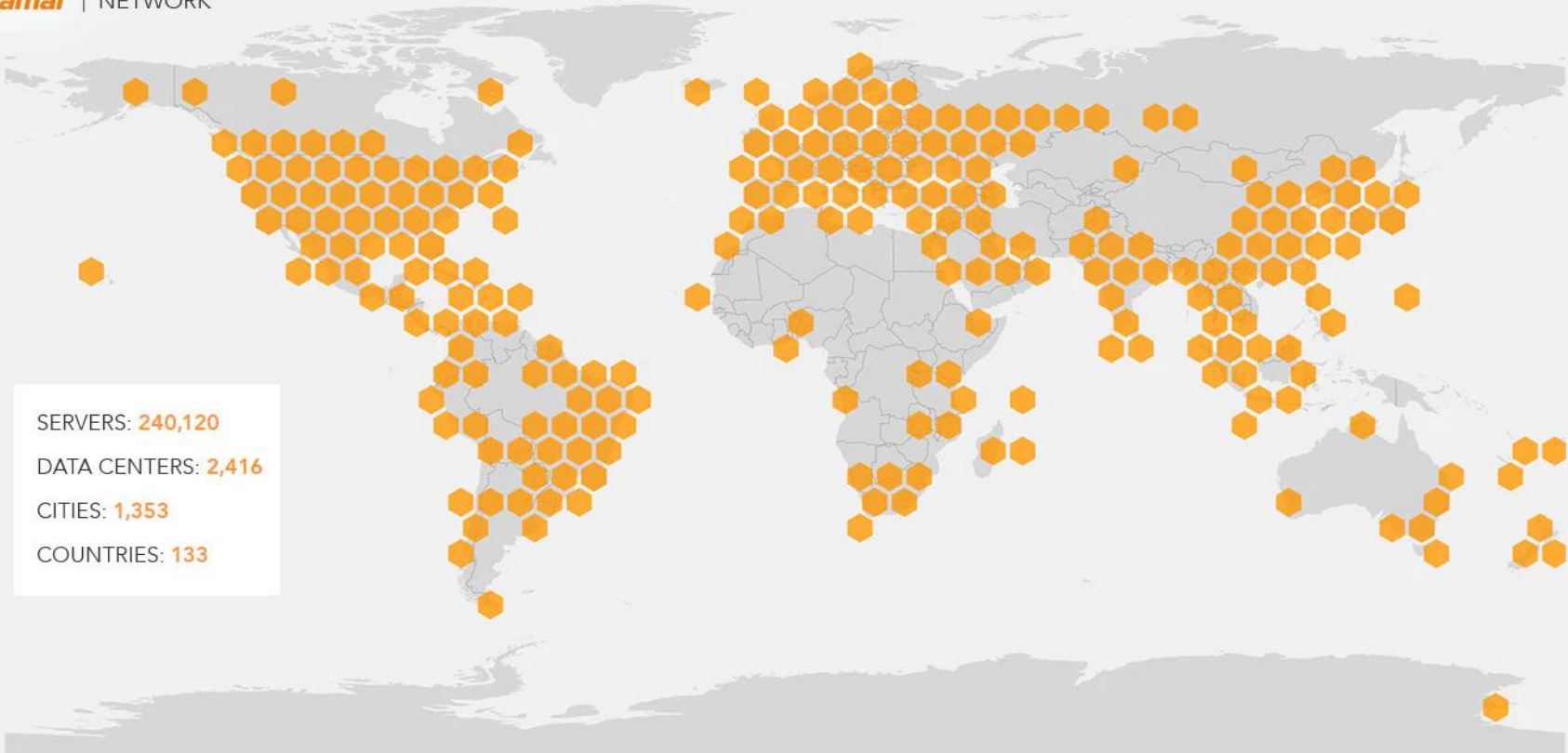


How Akamai works: *Cache hit*



Akamai infrastructure

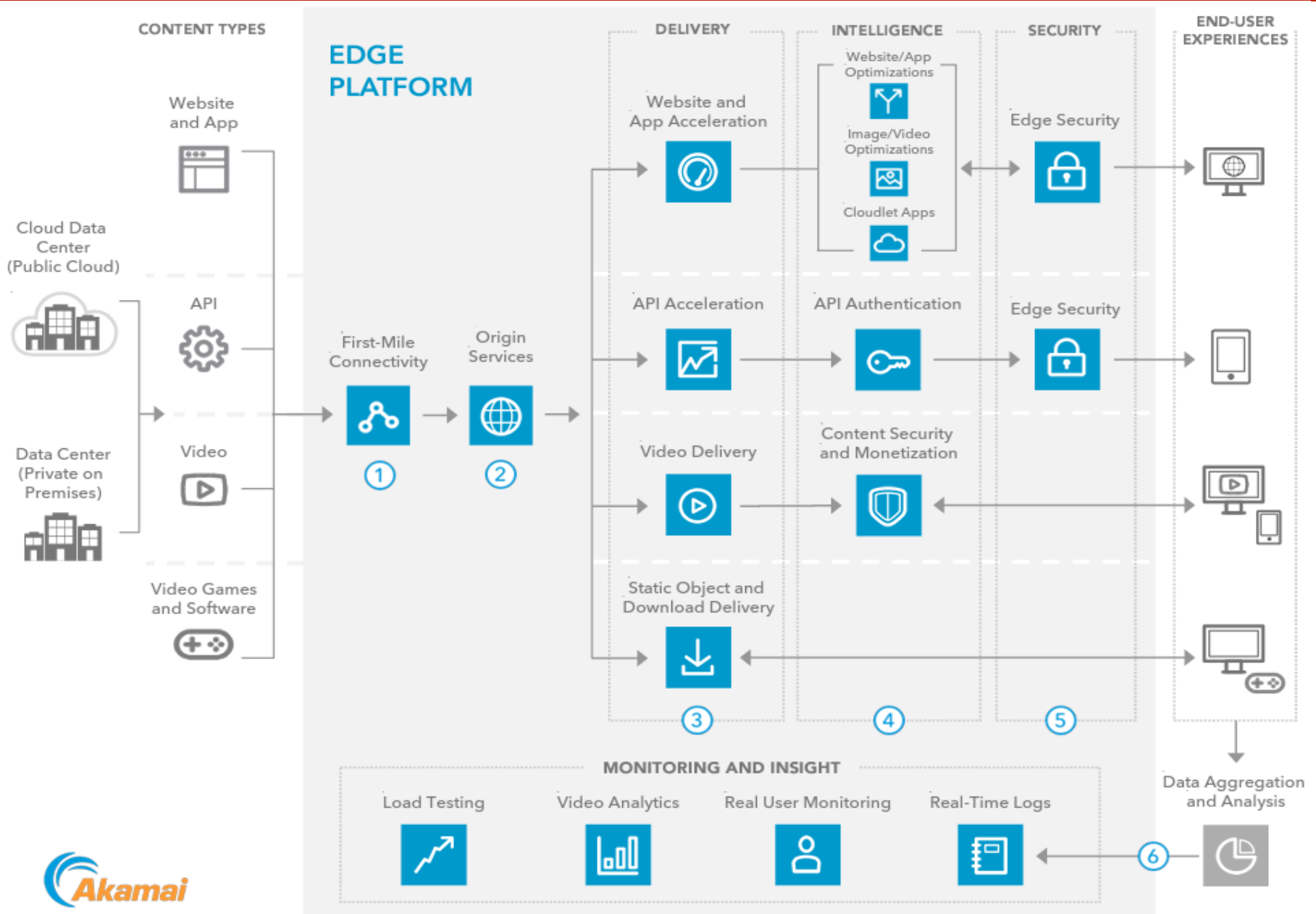
 Akamai | NETWORK



With a similar infrastructure, Akamai can offer CDN but even many other services



Content delivery (1/2)



Content delivery (2/2)

For many organizations, digital properties drive their business. Regardless of content type — websites, applications, APIs, video, large downloads—Akamai's set of content delivery products are designed to deliver powerful digital experiences to every user, across all locations, devices, or networks.

1. Access Akamai's cloud interconnects for a highly reliable, secure, and economical data transfer entry point from a public or private cloud. Customers in search of added assurance will connect directly to Akamai
 2. Store, ingest, or cache content away from cloud or on-premises origins, improving origin protection and offload, reducing egress costs, and minimizing infrastructure costs. Further optimize request routing using highly scalable load balancing
 3. Deliver flawless digital experiences to online audiences across any use case, at scale — regardless of location, device, or network. Each solution is built on the Akamai Intelligent Edge Platform and provides leading reliability, scale, reach, offload, and performance
 4. Further optimize end-user experiences and offload complex functions from origin infrastructure using the power and intelligence of Akamai's edge servers
 5. **Intelligent security starts at the edge:** protect websites, applications, APIs, and users with Akamai security solutions.
 6. Monitor delivery performance and analyze insights. Enhance proactive support through a managed service. Implement analytics engines for scalable, real-time load testing.
- Illustrate the relationship of performance and user engagement affecting your business with easy visualization.**

Proximity and mapping

- **Equivalence classes of IP addresses**
 - IP addresses experiencing similar performance
 - Quantify how well they connect to each other
- **Collect and combine measurements**
 - Ping, traceroute, BGP, server logs (>100 TB logs per day)
 - Network latency, loss, and connectivity
- **Map each IP class to a preferred server cluster**
 - Based on performance, cluster health, etc.
 - Updated roughly every minute
- **Map client request to a server in the cluster**
 - Load balancer selects a specific server (to maximize the cache hit rate, to improve load sharing)



A global view allows Akamai to adapt to failures (*network level*)

- **Failed path to customer's origin server**
 - Route packets through an intermediate node
- **Bad Internet routes**
 - Overlay routing through an intermediate server
- **Packet loss**
 - Sending redundant data over multiple paths
- **TCP connection set-up/teardown**
 - Pools of persistent connections
- **TCP congestion window and round-trip time**
 - Estimates based on network latency measurements

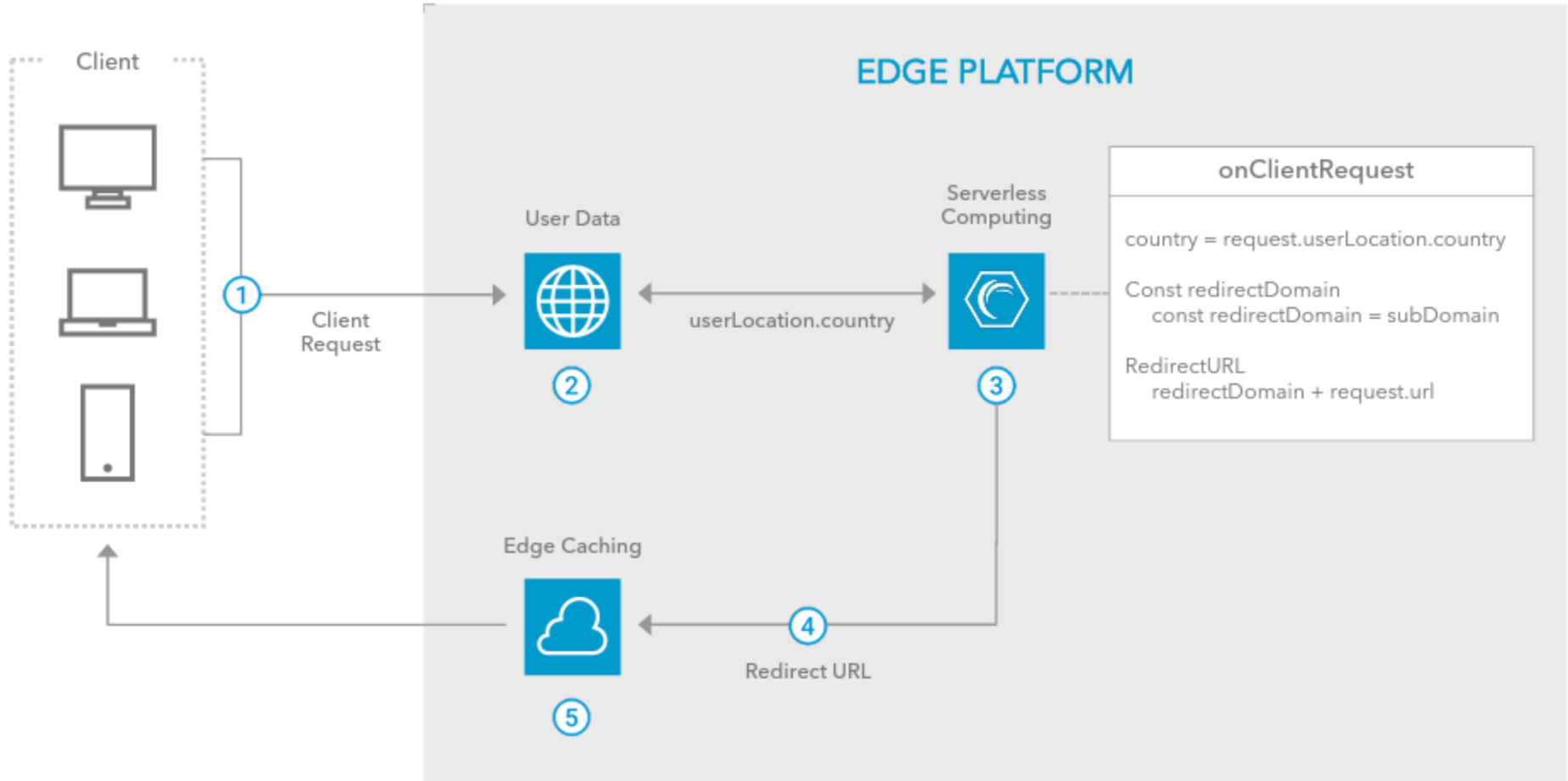


A global view allows Akamai to adapt to failures (*server level*)

- **Failing hard drive on a server**
 - Suspends after finishing “in progress” requests
- **Failed server**
 - Another server takes over for the IP address
 - Low-level map updated quickly
- **Failed cluster**
 - High-level map updated quickly



Geolocation and Redirect (1/2)



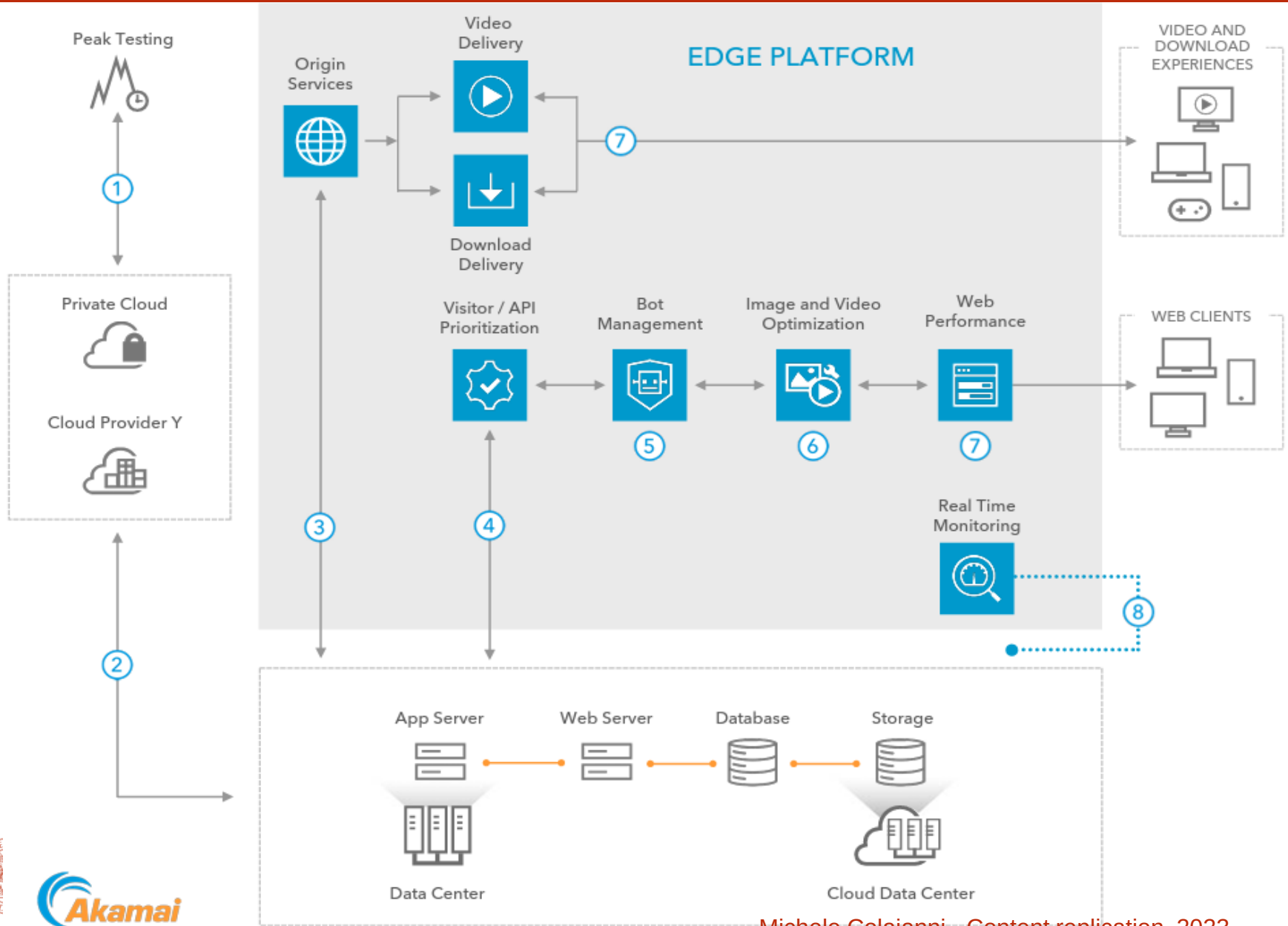
Geolocation and Redirect (2/2)

It is possible to leverage user geolocation data at the edge to enable applications to deliver location-specific personalized content to users

1. When interacting with a Web site, clients connect through an optimized Akamai edge server
2. The user's location is determined by **EdgeScape** that is, a geo lookup database built into the Akamai Intelligent Edge Platform
3. Serverless computing executes customer JavaScript leveraging client location data to:
 - Top-level domain based on the request's country of origin
 - Constructs domain based on the location of the request
 - Redirect URL returns the new domain in the response
4. The edge server provides a redirect URL to the user that requested the content
5. This compute request can then be cached to provide better performance in the future



Scaling for peaks (1/2)



Scaling for peaks (2/2)

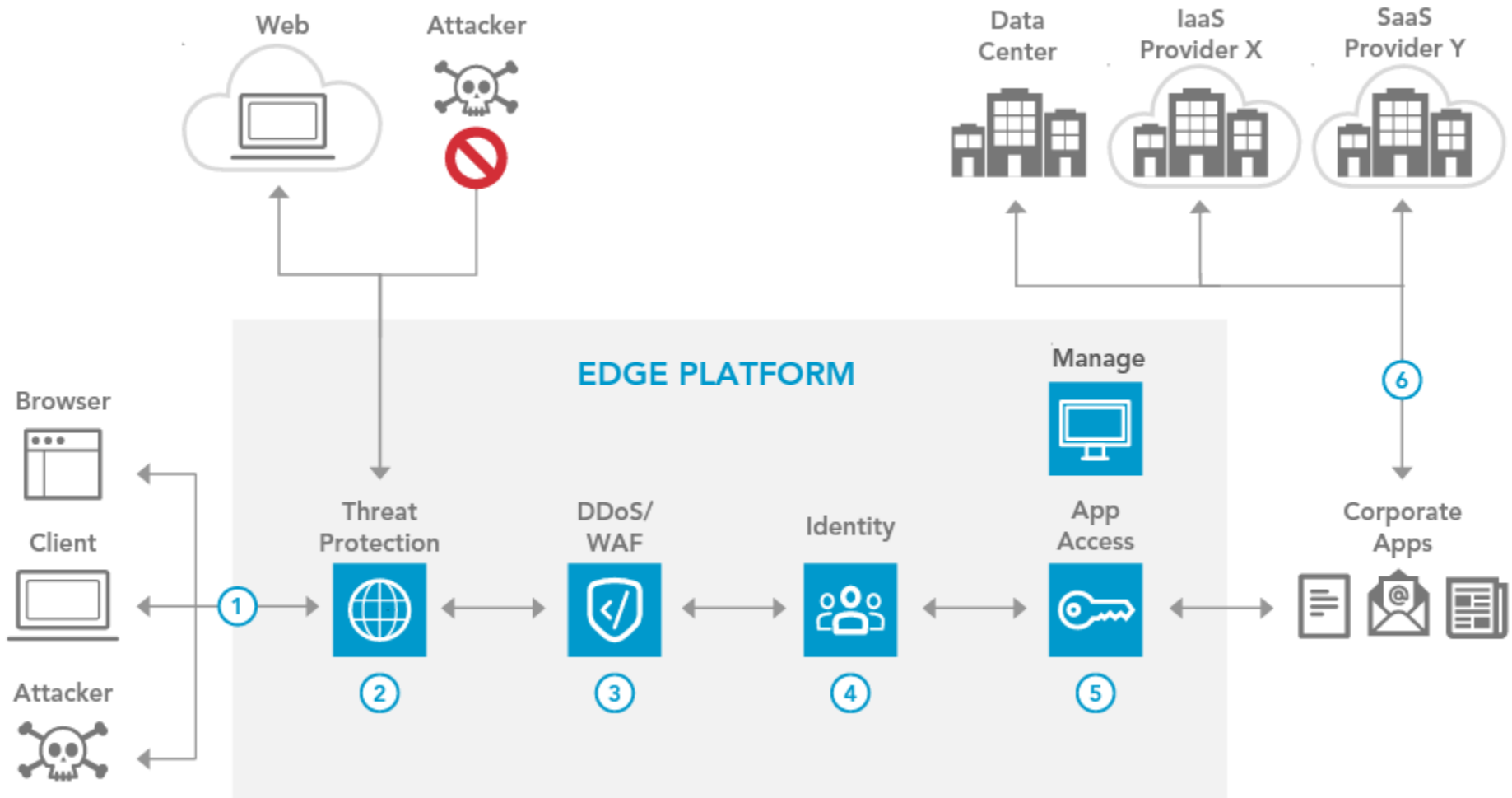
Complex modern Web applications have too many moving parts to scale easily. Akamai can help you prepare for a spike in demand and optimize resources and offloading when it hits —so users can have a good experience even during periods of extreme back-end load.

1. Peak testing generates traffic from a cloud testing platform that can comprise multiple different private and public clouds
2. Simulating traffic at peak scale with real-world conditions helps identify potential performance issues and breakpoints in the application environment before critical peak events
3. Store, ingest, or cache content away from cloud or on-premises origins, improving origin protection and offload, reducing egress/infrastructure costs
4. Visitor prioritization can help manage the user experience during periods of extremely heavy back-end load
5. Bot management helps shape bot traffic during peak periods to reduce load and maintain performance for human users
6. Image and video optimization delivers visual media with maximum quality and performance at the lowest possible bytes and with intelligent caching
7. The Akamai Intelligent Edge Platform delivers great user experiences and optimizes infrastructure offload as it scales to meet peak traffic demand

8. Real user monitoring, analytics and real-time logs help identify performance issues across a heterogeneous application environment in real time as it scales.



Zero Trust Architecture (1/2)



Zero Trust Architecture (2/2)

A Zero Trust security architecture minimizes the risk of malicious actors penetrating the perimeter, moving laterally, and exfiltrating data. Based on **least privilege** and **default deny**, Zero Trust lets you protect users and provide access through a single set of security and access controls, even as you scale finite resources to the needs of the business.

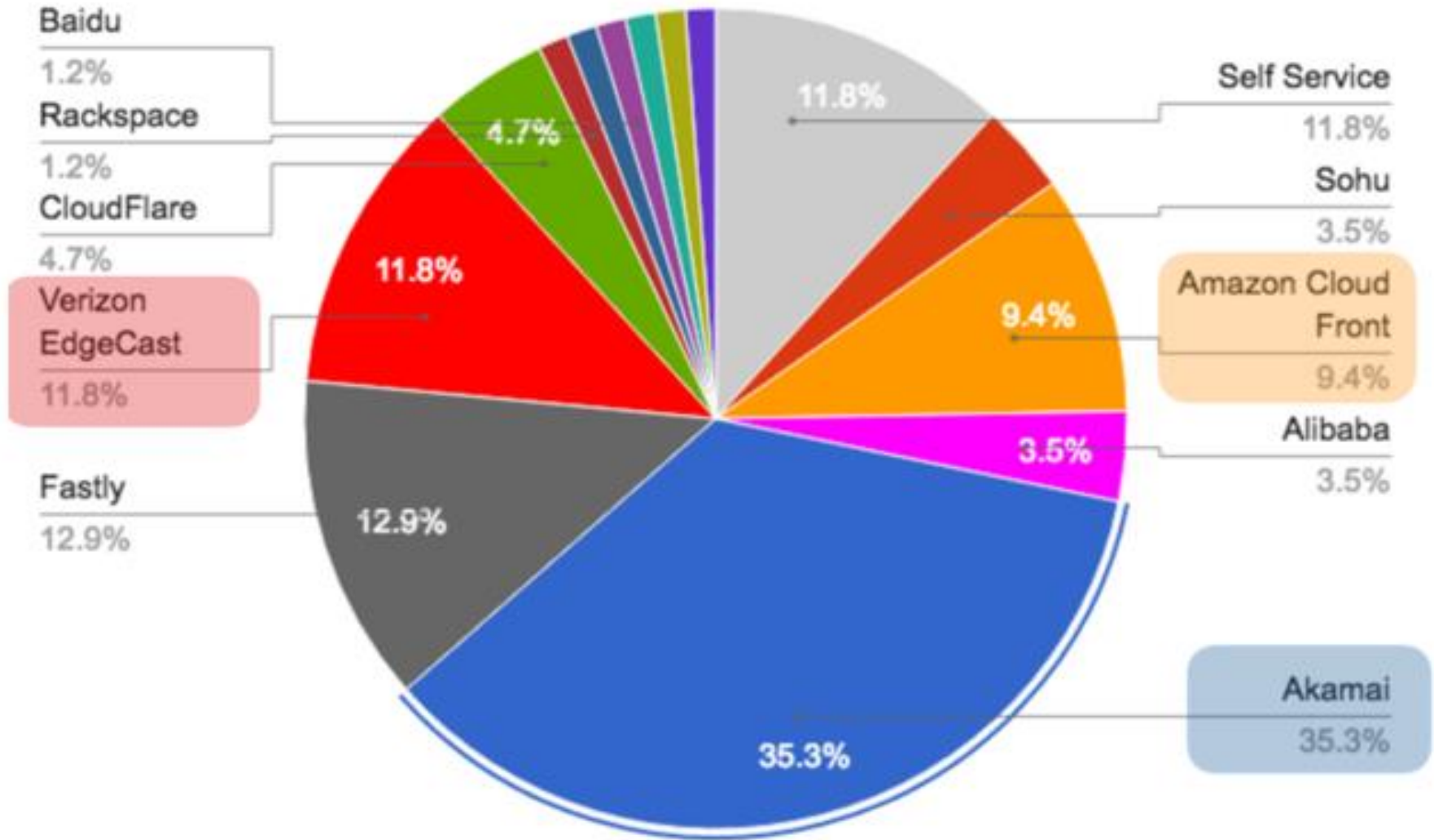
1. Users access corporate applications and the Web through the Akamai Intelligent Edge Platform
2. Threat protection defends users from malware, phishing, and malicious Web content, while providing visibility to the enterprise
3. For corporate applications, edge servers automatically drop network-layer DDoS attacks and inspect Web requests to block malicious threats like SQL injections, XSS, and RFI (Remote File Inclusion) script attacks
4. Identity of the user is established using on-premises, cloud-based, or Akamai identity stores
5. Based on the user's identity and other security signals, access is provided only to required applications, and not the entire corporate network
6. The Akamai Intelligent Edge Platform routes authorized and authenticated users to the relevant corporate applications

Total CDN industry

- 2014 \$2.78B
 - 2015: \$3.15B
 - 2016: \$3.61B
 - 2017: \$7.33B
 - ...
 - 2023: \$29.48B
- On average
 - **Data pricing:** customers doing 20PB a month, paying from \$0.005 to \$0.008 per GB
 - **Mbps pricing:** customers doing over 700 Mbps, paying from \$0.80 to \$1.10 per Mbps

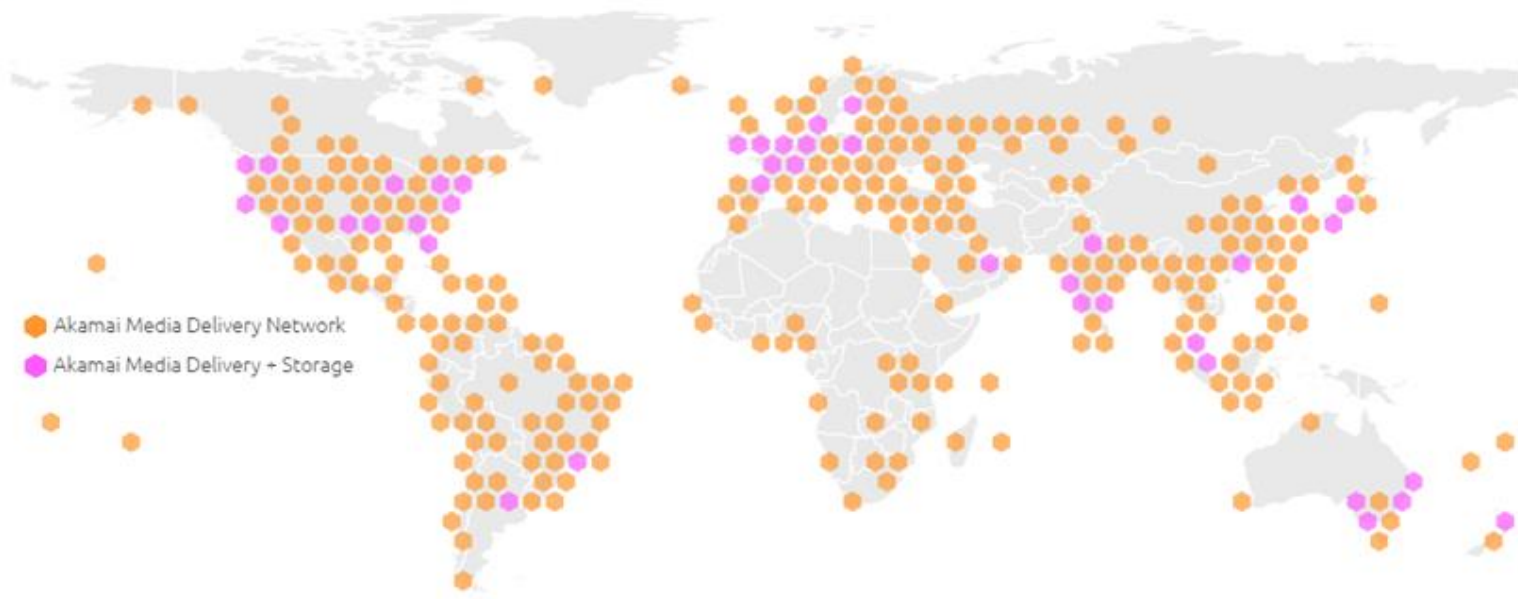


Market share



Akamai

1



Fastly

2

